

SUPERPOSITION IN DISJOINT VARIABLES FOR THE INFINITY LAPLACE EQUATION: BLUEPRINT OF THE LEAN FORMALIZATION

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ABSTRACT. This document collects the natural-language statements and detailed proofs that accompany the Lean 4 formalization of the main results of Q. Liu, J. J. Manfredi and X. Zhou, *Superposition property in disjoint variables for the infinity Laplace equation*, RIMS Kôkyûroku Bessatsu B99 (2026), 19–34, arXiv:2508.17106. Each subsection corresponds to the Lean declaration of the same name in the directory `Superposition/`; cross-references point to the results that a proof uses. Items appear in dependency order.

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1. DEFINITIONS

1.1. ConeComparisonAbove.

Definition 1.1 (Comparison with cones from above). Let $(\mathbf{X}, d_{\mathbf{X}})$ be a metric space, $\Omega \subseteq \mathbf{X}$, and let u be a real-valued function defined (at least) on Ω ; only the values of u on Ω matter below. We say that u *satisfies comparison with cones from above in Ω* (with respect to $d_{\mathbf{X}}$) if both of the following hold.

- (1) u is locally bounded from above in Ω : for every $x \in \Omega$ there is a neighbourhood t of x in $\mathbf{X}$ such that u is bounded from above on $t \cap \Omega$.
- (2) For any $\hat{x} \in \Omega$, any $c \in \mathbb{R}$, any $\kappa \geq 0$ and any bounded open set $\mathcal{O} \subset\subset \Omega$ (that is, $\mathcal{O} \subseteq \mathbf{X}$ is open and bounded, its closure $\overline{\mathcal{O}}$ is compact, and $\overline{\mathcal{O}} \subseteq \Omega$) with $\hat{x} \in \Omega \setminus \mathcal{O}$, the condition

$$u \leq \phi \quad \text{on } \partial\mathcal{O}$$

for the cone function $\phi = c + \kappa d_{\mathbf{X}}(\hat{x}, \cdot)$ implies that

$$u \leq \phi \quad \text{in } \overline{\mathcal{O}}.$$

In the paper this notion is used when $(\mathbf{X}, d_{\mathbf{X}})$ is a proper geodesic space and $\Omega \subsetneq \mathbf{X}$ is a domain; these standing assumptions are imposed as explicit hypotheses in the theorems that use this definition.

1.2. ConeComparisonBelow.

Definition 1.2 (Comparison with cones from below). Let $(\mathbf{X}, d_{\mathbf{X}})$ be a metric space, $\Omega \subseteq \mathbf{X}$, and let u be a real-valued function defined (at least) on Ω . We say that u *satisfies comparison with cones from below in Ω* if $-u$ satisfies comparison with cones from above in Ω in the sense of [ConeComparisonAbove](#). Unfolding that definition, this says: u is locally bounded from below in Ω (every $x \in \Omega$ has a neighbourhood t with u bounded from below on $t \cap \Omega$), and for any $\hat{x} \in \Omega$, $c \in \mathbb{R}$, $\kappa \geq 0$ and bounded open $\mathcal{O} \subset\subset \Omega$ with $\hat{x} \in \Omega \setminus \mathcal{O}$, the condition $-u \leq c + \kappa d_{\mathbf{X}}(\hat{x}, \cdot)$ on $\partial\mathcal{O}$ implies $-u \leq c + \kappa d_{\mathbf{X}}(\hat{x}, \cdot)$ in $\overline{\mathcal{O}}$.

1.3. ConeComparisonBoth.

Definition 1.3 (Comparison with cones from both sides). Let $(\mathbf{X}, d_{\mathbf{X}})$ be a metric space, $\Omega \subseteq \mathbf{X}$, and let u be a real-valued function defined (at least) on Ω . We say that u *satisfies comparison with cones (from both sides) in Ω* if u satisfies comparison with cones from above in Ω ([ConeComparisonAbove](#)) and u satisfies comparison with cones from below in Ω ([ConeComparisonBelow](#)). In particular such a u is locally bounded (from above and from below) in Ω .

1.4. EuclidFst.

Definition 1.4 (First coordinate block of $\mathbb{R}^{n+m}$). Let $n, m \geq 0$ and identify $\mathbb{R}^{n+m}$ with Euclidean space (Lean: `EuclideanSpace  $\mathbb{R}$  (Fin (n+m))`), with coordinates $z = (z_1, \dots, z_{n+m})$. The *first block* of z is the vector

$$z' = \text{fst}(z) = (z_1, \dots, z_n) \in \mathbb{R}^n,$$

i.e. $\text{fst}(z)_i = z_i$ for $1 \leq i \leq n$ (Lean: `coordinate Fin.castAdd m i`). Together with the second block $\text{snd}(z) = (z_{n+1}, \dots, z_{n+m}) \in \mathbb{R}^m$ this realizes the identification $\mathbb{R}^{n+m} = \mathbb{R}^n \times \mathbb{R}^m$ by concatenation of coordinates, $z \leftrightarrow (\text{fst } z, \text{snd } z)$, used for the product $\mathcal{U} \times \mathcal{V} \subseteq \mathbb{R}^n \times \mathbb{R}^m$ of paper Theorem `infsuper`, where the point (x, y) is written z with $\text{fst } z = x$.

1.5. EuclidHessian.

Definition 1.5 (Hessian matrix). Let $n \geq 0$ and identify $\mathbb{R}^n$ with Euclidean space (Lean: `EuclideanSpace  $\mathbb{R}$  (Fin n)`), with standard basis $e_1, \dots, e_n$. For $\psi : \mathbb{R}^n \rightarrow \mathbb{R}$ let $D\psi(x)$ denote the Fréchet derivative of ψ at x (a linear map $\mathbb{R}^n \rightarrow \mathbb{R}$), with the convention that $D\psi(x) = 0$ when ψ is not differentiable at x (Mathlib's `fderiv`); likewise $D^2\psi(x) = D(D\psi)(x)$ is the Fréchet derivative at x of the map $y \mapsto D\psi(y)$ (with the same convention), so that $D^2\psi(x)(h)(k) \in \mathbb{R}$ for $h, k \in \mathbb{R}^n$.

The *Hessian matrix* of ψ at x is the $n \times n$ real matrix $\nabla^2\psi(x)$ with entries

$$(\nabla^2\psi(x))_{ij} = D^2\psi(x)(e_i)(e_j), \quad 1 \leq i, j \leq n.$$

When ψ is C^2 in a neighbourhood of x , this is the usual matrix of second partial derivatives $\partial_i \partial_j \psi(x)$ appearing as $\nabla^2\psi$ in the paper.

1.6. EuclidSnd.

Definition 1.6 (Second coordinate block of $\mathbb{R}^{n+m}$). Let $n, m \geq 0$ and identify $\mathbb{R}^{n+m}$ with Euclidean space (Lean: `EuclideanSpace  $\mathbb{R}$  (Fin (n+m))`), with coordinates $z = (z_1, \dots, z_{n+m})$. The *second block* of z is the vector

$$z'' = \text{snd}(z) = (z_{n+1}, \dots, z_{n+m}) \in \mathbb{R}^m,$$

i.e. $\text{snd}(z)_j = z_{n+j}$ for $1 \leq j \leq m$ (Lean: `coordinate Fin.natAdd n j`). Together with the first block $\text{fst}(z) = (z_1, \dots, z_n)$ this realizes the identification $\mathbb{R}^{n+m} = \mathbb{R}^n \times \mathbb{R}^m$ by concatenation of coordinates, $z \leftrightarrow (\text{fst } z, \text{snd } z)$, as in paper Theorem `infsuper`, where the point (x, y) has $\text{snd } z = y$.

1.7. InfLaplaceOperator.

Definition 1.7 (The infinity-Laplace operator). Let $n \geq 0$. The operator $L_\infty : \mathbb{R}^n \times \mathbb{R}^{n \times n} \rightarrow \mathbb{R}$ is

$$L_\infty(\xi, X) = -\langle X\xi, \xi \rangle,$$

where $X\xi$ is the matrix-vector product and $\langle \cdot, \cdot \rangle$ the Euclidean inner product. Thus for ψ twice differentiable at x , $L_\infty(\nabla\psi(x), \nabla^2\psi(x)) = -\langle \nabla^2\psi(x)\nabla\psi(x), \nabla\psi(x) \rangle = -\Delta_\infty\psi(x)$, and a viscosity sub/supersolution of $-\Delta_\infty u = f$ in the paper's sense (Definition `def-vis` with $L(x, \xi, X) = -\langle X\xi, \xi \rangle - f(x)$, paper lines following `def-vis`) is a viscosity sub/supersolution of $L_\infty(\nabla u, \nabla^2 u) = f$.

1.8. IsDegenerateElliptic.

Definition 1.8 (Elliptic operator, paper eq. (ellip)). Let $n \geq 0$, let $\mathbb{R}^{n \times n}$ be the real $n \times n$ matrices and $\mathbb{S}^n \subseteq \mathbb{R}^{n \times n}$ the symmetric ones. For $X, Y \in \mathbb{R}^{n \times n}$ write $X \geq Y$ if $X - Y$ is positive semidefinite, i.e. $X - Y$ is symmetric and $\langle (X - Y)h, h \rangle \geq 0$ for all $h \in \mathbb{R}^n$.

An operator $L : \mathbb{R}^n \times \mathbb{R}^{n \times n} \rightarrow \mathbb{R}$ is (*degenerate*) *elliptic* if

$$L(\xi, X) \leq L(\xi, Y) \quad \text{for all } \xi \in \mathbb{R}^n \text{ and all } X, Y \in \mathbb{S}^n \text{ with } X \geq Y.$$

Only symmetric matrices enter this condition, so it is exactly the paper's ellipticity condition (ellip) for an operator $L(\xi, X)$ on $\mathbb{R}^n \times \mathbb{S}^n$ (the values of L at non-symmetric matrices play no role).

1.9. IsGeodesicSpace.

Definition 1.9 (Geodesic space). A metric space $(\mathbf{X}, d_{\mathbf{X}})$ is a *geodesic space* if for any two points $x, y \in \mathbf{X}$ there is a map $\gamma : \mathbb{R} \rightarrow \mathbf{X}$ with $\gamma(0) = x$, $\gamma(d_{\mathbf{X}}(x, y)) = y$, and

$$d_{\mathbf{X}}(\gamma(s), \gamma(t)) = |s - t| \quad \text{for all } s, t \in [0, d_{\mathbf{X}}(x, y)],$$

that is, x and y are joined by a geodesic segment (an isometric copy of the interval $[0, d_{\mathbf{X}}(x, y)]$). (The values of γ outside $[0, d_{\mathbf{X}}(x, y)]$ play no role.)

1.10. IsOpenDomain.

Definition 1.10 (Domain). Let $\mathbf{X}$ be a topological space. A subset $\Omega \subseteq \mathbf{X}$ is a *domain* if it is open, nonempty and connected.

1.11. IsSubadditiveOperator.

Definition 1.11 (Subadditive operator, paper eq. (sub-add)). Let $n \geq 0$ and let $\mathbb{S}^n$ denote the symmetric real $n \times n$ matrices. An operator $L : \mathbb{R}^n \times \mathbb{R}^{n \times n} \rightarrow \mathbb{R}$ is *subadditive* if

$$L(\xi + \eta, X + Y) \leq L(\xi, X) + L(\eta, Y) \quad \text{for all } \xi, \eta \in \mathbb{R}^n \text{ and all } X, Y \in \mathbb{S}^n.$$

This is the paper's condition (sub-add); only symmetric matrices enter it.

1.12. IsViscositySubsolution.

Definition 1.12 (Viscosity subsolution of $L(\nabla u, \nabla^2 u) = f$, paper Def. def-vis). Let $n \geq 0$, $\Omega \subseteq \mathbb{R}^n$, $L : \mathbb{R}^n \times \mathbb{R}^{n \times n} \rightarrow \mathbb{R}$ and $f, u : \mathbb{R}^n \rightarrow \mathbb{R}$. We say that u is a *viscosity subsolution* of $L(\nabla u, \nabla^2 u) = f$ in Ω if

- (i) u is upper semicontinuous on Ω (relative to Ω : for every $x \in \Omega$ and $t > u(x)$ there is a neighbourhood U of x with $u < t$ on $U \cap \Omega$), and
- (ii) whenever $x_0 \in \Omega$ and $\psi : \mathbb{R}^n \rightarrow \mathbb{R}$ is C^2 on Ω (Lean `ContDiffOn` $\mathbb{R} \ 2 \ \psi \ \Omega$) are such that $u - \psi$ attains a local maximum at x_0 (i.e. $u(x) - \psi(x) \leq u(x_0) - \psi(x_0)$ for all x in some neighbourhood of x_0 in $\mathbb{R}^n$), we have

$$L(\nabla \psi(x_0), \nabla^2 \psi(x_0)) \leq f(x_0),$$

where $\nabla \psi(x_0)$ is the gradient (the vector representing $D\psi(x_0)$ via the inner product) and $\nabla^2 \psi(x_0)$ is the Hessian matrix of `EuclidHessian`.

For an open set Ω (the only case used), neighbourhoods of $x_0 \in \Omega$ may be taken inside Ω , the values of u, f, ψ outside Ω are irrelevant, and this is exactly the paper's Definition def-vis of a viscosity subsolution $u \in USC(\Omega)$ of $L(\nabla u, \nabla^2 u) = f$, i.e. of $\tilde{L}(x, \xi, X) = L(\xi, X) - f(x) = 0$, with test functions $\psi \in C^2(\Omega)$.

1.13. IsViscositySupersolution.

Definition 1.13 (Viscosity supersolution of $L(\nabla u, \nabla^2 u) = f$, paper Def. def-vis). Let $n \geq 0$, $\Omega \subseteq \mathbb{R}^n$, $L : \mathbb{R}^n \times \mathbb{R}^{n \times n} \rightarrow \mathbb{R}$ and $f, u : \mathbb{R}^n \rightarrow \mathbb{R}$. We say that u is a *viscosity supersolution* of $L(\nabla u, \nabla^2 u) = f$ in Ω if

- (i) u is lower semicontinuous on Ω (relative to Ω : for every $x \in \Omega$ and $t < u(x)$ there is a neighbourhood U of x with $u > t$ on $U \cap \Omega$), and
- (ii) whenever $x_0 \in \Omega$ and $\psi : \mathbb{R}^n \rightarrow \mathbb{R}$ is C^2 on Ω (Lean `ContDiffOn` $\mathbb{R} \ 2 \ \psi \ \Omega$) are such that $u - \psi$ attains a local minimum at x_0 (i.e. $u(x) - \psi(x) \geq u(x_0) - \psi(x_0)$ for all x in some neighbourhood of x_0 in $\mathbb{R}^n$), we have

$$f(x_0) \leq L(\nabla \psi(x_0), \nabla^2 \psi(x_0)),$$

where $\nabla \psi(x_0)$ is the gradient and $\nabla^2 \psi(x_0)$ the Hessian matrix of `EuclidHessian`.

For an open Ω this is exactly the paper's Definition def-vis of a viscosity supersolution $u \in LSC(\Omega)$ of $L(\nabla u, \nabla^2 u) = f$, i.e. of $\tilde{L}(x, \xi, X) = L(\xi, X) - f(x) = 0$ ($\tilde{L} \geq 0$ at test points), with test functions $\psi \in C^2(\Omega)$.

1.14. `IsViscositySolution`.

Definition 1.14 (Viscosity solution of $L(\nabla u, \nabla^2 u) = f$, paper Def. def-vis). Let $n \geq 0$, $\Omega \subseteq \mathbb{R}^n$, $L : \mathbb{R}^n \times \mathbb{R}^{n \times n} \rightarrow \mathbb{R}$ and $f, u : \mathbb{R}^n \rightarrow \mathbb{R}$. We say that u is a *viscosity solution* of $L(\nabla u, \nabla^2 u) = f$ in Ω if u is both a viscosity subsolution (`IsViscositySubsolution`) and a viscosity supersolution (`IsViscositySupersolution`) of $L(\nabla u, \nabla^2 u) = f$ in Ω . Since the first requires u to be upper and the second lower semicontinuous on Ω , u is then continuous on Ω , i.e. $u \in C(\Omega)$, matching the paper's definition of a viscosity solution as a function $u \in C(\Omega)$ that is both a viscosity subsolution and a viscosity supersolution.

1.15. `SupConvolution`.

Definition 1.15 (Sup-convolution). Let $n \geq 0$, $u : \mathbb{R}^n \rightarrow \mathbb{R}$, $K \subseteq \mathbb{R}^n$ and $\varepsilon \in \mathbb{R}$. The *sup-convolution* of u over K with parameter ε is the function $u_K^\varepsilon : \mathbb{R}^n \rightarrow \mathbb{R}$,

$$u_K^\varepsilon(x) = \sup_{y \in K} \left(u(y) - \frac{|x - y|^2}{2\varepsilon} \right),$$

where sup is the real supremum of the set $\{u(y) - |x - y|^2/(2\varepsilon) : y \in K\}$, with Lean's convention that the supremum of an empty set or of a set that is not bounded above is 0. (It is only used for $\varepsilon > 0$, K compact nonempty and u upper semicontinuous on K , where the supremum is a genuine, attained maximum.)

2. AUXILIARY RESULTS

2.1. `convex_exists_subgradient`.

Lemma 2.1 (Existence of subgradients). *Let $n \geq 0$ and let $F : \mathbb{R}^n \rightarrow \mathbb{R}$ be convex on $\mathbb{R}^n$. Then for every $x \in \mathbb{R}^n$ there is $q \in \mathbb{R}^n$ (a subgradient of F at x) such that*

$$F(x) + \langle q, y - x \rangle \leq F(y) \quad \text{for all } y \in \mathbb{R}^n.$$

Proof. Fix x . For $h \in \mathbb{R}^n$ and $t > 0$ put $D_h(t) = \frac{F(x+th) - F(x)}{t}$.

Monotonicity. For $0 < s < t$, $x + sh = \frac{s}{t}(x + th) + (1 - \frac{s}{t})x$, so convexity gives $F(x + sh) \leq \frac{s}{t}F(x + th) + (1 - \frac{s}{t})F(x)$, i.e. $F(x + sh) - F(x) \leq \frac{s}{t}(F(x + th) - F(x))$, i.e. $D_h(s) \leq D_h(t)$.

Lower bound. For $t > 0$, $x = \frac{1}{1+t}(x + th) + \frac{t}{1+t}(x - h)$, so $(1 + t)F(x) \leq F(x + th) + tF(x - h)$, i.e. $D_h(t) \geq F(x) - F(x - h)$.

Hence $\varphi(h) = \inf_{t>0} D_h(t)$ is a real number for every h , and $\varphi(h) \leq D_h(1) = F(x + h) - F(x)$.

φ is sublinear. (a) For $\lambda > 0$: $D_{\lambda h}(t) = \lambda D_h(\lambda t)$, and $t \mapsto \lambda t$ is a bijection of $(0, \infty)$, so $\varphi(\lambda h) = \lambda \varphi(h)$. (b) For h, k and $t > 0$: $x + t(h + k) = \frac{1}{2}(x + 2th) + \frac{1}{2}(x + 2tk)$, so $F(x + t(h + k)) \leq \frac{1}{2}F(x + 2th) + \frac{1}{2}F(x + 2tk)$, which rearranges to $D_{h+k}(t) \leq D_h(2t) + D_k(2t)$. Given $s_1, s_2 > 0$ take $t = \frac{1}{2} \min(s_1, s_2)$; by monotonicity $D_h(2t) \leq D_h(s_1)$ and $D_k(2t) \leq D_k(s_2)$, so $\varphi(h + k) \leq D_h(s_1) + D_k(s_2)$. Taking the infimum over s_1 and then over s_2 gives $\varphi(h + k) \leq \varphi(h) + \varphi(k)$.

Hahn–Banach. Apply the Hahn–Banach extension theorem (dominated version; `Mathlib.exists_extension_of_le_sublinear`) to the zero functional on the subspace $\{0\}$, which is dominated by φ there since $\varphi(0) = \inf_t D_0(t) = 0$. It yields a linear functional $\ell : \mathbb{R}^n \rightarrow \mathbb{R}$ with $\ell(h) \leq \varphi(h)$ for all h . Since $\mathbb{R}^n$ is a finite-dimensional inner product space, $\ell = \langle q, \cdot \rangle$ with $q = \sum_i \ell(e_i) e_i$.

Conclusion. For $y \in \mathbb{R}^n$ and $h = y - x$: $\langle q, y - x \rangle = \ell(h) \leq \varphi(h) \leq F(x + h) - F(x) = F(y) - F(x)$. $\square$

2.2. `convex_locally_lipschitz`.

Lemma 2.2 (Convex functions are locally Lipschitz with locally bounded subgradients). *Let $n \geq 0$, let $F : \mathbb{R}^n \rightarrow \mathbb{R}$ be convex on $\mathbb{R}^n$, $x_0 \in \mathbb{R}^n$ and $R \in \mathbb{R}$. Then there is $\Lambda \geq 0$ such that*

- (i) $|F(y) - F(y')| \leq \Lambda |y - y'|$ for all $y, y' \in \overline{B}(x_0, R)$, and
- (ii) $|q| \leq \Lambda$ whenever $y \in \overline{B}(x_0, R)$ and $q \in \mathbb{R}^n$ satisfies $F(y) + \langle q, w - y \rangle \leq F(w)$ for all $w \in \mathbb{R}^n$.

Proof. If $R < 0$ the ball is empty and $\Lambda = 0$ works; assume $R \geq 0$ and put $\rho = R + 1$.

Upper bound on $\bar{B}(x_0, \rho)$. Let $s = (n + 1)\rho$ and $M = \max\{F(x_0), F(x_0 \pm se_i) : 1 \leq i \leq n\}$. Let $|h| \leq \rho$. By Cauchy–Schwarz $\sum_i |h_i| \leq \sqrt{n}|h| \leq s$. Let $\sigma_i = 1$ if $h_i \geq 0$ and $\sigma_i = -1$ otherwise, $\lambda_i = |h_i|/s$ and $\lambda_0 = 1 - \sum_i \lambda_i \geq 0$. Then $\lambda_0 + \sum_i \lambda_i = 1$ and

$$x_0 + h = \lambda_0 x_0 + \sum_{i=1}^n \lambda_i (x_0 + \sigma_i s e_i),$$

so by Jensen’s inequality for the convex F (finite convex combinations), $F(x_0 + h) \leq \lambda_0 F(x_0) + \sum_i \lambda_i F(x_0 + \sigma_i s e_i) \leq M$.

Lower bound. For $y \in \bar{B}(x_0, \rho)$ also $y' = 2x_0 - y \in \bar{B}(x_0, \rho)$ and $x_0 = \frac{1}{2}y + \frac{1}{2}y'$, so $F(x_0) \leq \frac{1}{2}F(y) + \frac{1}{2}F(y') \leq \frac{1}{2}F(y) + \frac{1}{2}M$, i.e. $F(y) \geq 2F(x_0) - M$. Hence $|F| \leq C$ on $\bar{B}(x_0, \rho)$ with $C = \max(|M|, |2F(x_0) - M|)$. Put $\Lambda = 2C \geq 0$.

(i). Let $y, y' \in \bar{B}(x_0, R)$ with $d = |y - y'| > 0$ (for $d = 0$ there is nothing to show), let $e = (y' - y)/d$ and $w = y' + e$. Then $|w - x_0| \leq R + 1 = \rho$, and $y' = \lambda w + (1 - \lambda)y$ with $\lambda = \frac{d}{d+1}$ (indeed $y + de = y'$ and $w = y + (d + 1)e$). Convexity gives $F(y') \leq \lambda F(w) + (1 - \lambda)F(y)$, i.e. $F(y') - F(y) \leq \lambda(F(w) - F(y)) \leq \frac{d}{d+1} 2C \leq \Lambda d$. Exchanging y, y' gives $|F(y) - F(y')| \leq \Lambda |y - y'|$.

(ii). Let $y \in \bar{B}(x_0, R)$ and q as in (ii); if $q = 0$ there is nothing to show, so let $q \neq 0$. Then $w = y + q/|q| \in \bar{B}(x_0, \rho)$ and $F(w) \geq F(y) + \langle q, q/|q| \rangle = F(y) + |q|$, so $|q| \leq F(w) - F(y) \leq 2C = \Lambda$. □

2.3. convex_prox_map.

Lemma 2.3 (The proximal map of a convex function). *Let $n \geq 0$ and let $F : \mathbb{R}^n \rightarrow \mathbb{R}$ be convex on $\mathbb{R}^n$. Call $q \in \mathbb{R}^n$ a subgradient of F at x if $F(x) + \langle q, w - x \rangle \leq F(w)$ for all $w \in \mathbb{R}^n$. Then there is a map $P : \mathbb{R}^n \rightarrow \mathbb{R}^n$ such that*

- (i) *for every z , $z - P(z)$ is a subgradient of F at $P(z)$;*
- (ii) *P is 1-Lipschitz: $|P(z) - P(z')| \leq |z - z'|$ for all z, z' ;*
- (iii) *if q is a subgradient of F at x , then $P(x + q) = x$.*

Proof. Monotonicity. If q is a subgradient at x and q' a subgradient at x' , then adding $F(x') \geq F(x) + \langle q, x' - x \rangle$ and $F(x) \geq F(x') + \langle q', x - x' \rangle$ gives $\langle q - q', x - x' \rangle \geq 0$.

Key estimate. Suppose $z - x$ is a subgradient at x and $z' - x'$ is a subgradient at x' . By monotonicity $\langle (z - x) - (z' - x'), x - x' \rangle \geq 0$, i.e. $\langle z - z', x - x' \rangle \geq |x - x'|^2$. By Cauchy–Schwarz $|x - x'|^2 \leq |z - z'| |x - x'|$, hence $|x - x'| \leq |z - z'|$ (trivially if $x = x'$). (K)

Existence of a point x with $z - x$ a subgradient at x . Fix z and let $G(y) = F(y) + \frac{1}{2}|y - z|^2$. By [convex_exists_subgradient](#) there is a subgradient q_0 of F at z , so $F(y) \geq F(z) + \langle q_0, y - z \rangle \geq F(z) - |q_0| |y - z|$ and

$$G(y) \geq G(z) + |y - z|(\frac{1}{2}|y - z| - |q_0|) > G(z) \quad \text{whenever } |y - z| > 2|q_0|.$$

F is continuous: for every y_0 , [convex_locally_lipschitz](#)(i) with centre y_0 and radius 1 gives Λ with $|F(y) - F(y_0)| \leq \Lambda |y - y_0|$ for $|y - y_0| \leq 1$. So G is continuous and attains its minimum over the compact nonempty ball $\bar{B}(z, 2|q_0|)$ at some x ; since $G(x) \leq G(z) < G(y)$ for y outside that ball, x minimizes G over all of $\mathbb{R}^n$. Now let $w \in \mathbb{R}^n$ and $t \in (0, 1)$. By minimality and convexity of F ,

$$F(x) + \frac{1}{2}|x - z|^2 \leq F(x + t(w - x)) + \frac{1}{2}|x - z + t(w - x)|^2 \leq (1 - t)F(x) + tF(w) + \frac{1}{2}|x - z|^2 + t\langle x - z, w - x \rangle + \frac{t^2}{2}|w - x|^2.$$

Cancelling and dividing by t : $0 \leq F(w) - F(x) + \langle x - z, w - x \rangle + \frac{t}{2}|w - x|^2$. Letting $t \rightarrow 0^+$ gives $F(x) + \langle z - x, w - x \rangle \leq F(w)$. So $z - x$ is a subgradient at x .

Definition of P and (i). For each z choose such an x and call it $P(z)$; (i) holds by construction.

(ii). Apply (K) to $x = P(z)$, $x' = P(z')$ (admissible by (i)): $|P(z) - P(z')| \leq |z - z'|$.

(iii). Let q be a subgradient at x and $z = x + q$. Then $z - x = q$ is a subgradient at x , and by (i) $z - P(z)$ is a subgradient at $P(z)$. Applying (K) with $z' = z$ gives $|x - P(z)| \leq 0$, i.e. $P(x + q) = x$. □

2.4. convex_second_order_of_subgradient_expansion.

Lemma 2.4 (Second-order expansion from a first-order expansion of subgradients). *Let $n \geq 0$, let $F : \mathbb{R}^n \rightarrow \mathbb{R}$ be convex on $\mathbb{R}^n$, and call q a subgradient of F at y if $F(y) + \langle q, w - y \rangle \leq F(w)$ for all $w \in \mathbb{R}^n$. Let $x, a \in \mathbb{R}^n$ and let A_0 be a real $n \times n$ matrix (not necessarily symmetric) such that for every $\kappa > 0$ there is $\tau > 0$ with*

$$|q - a - A_0(y - x)| \leq \kappa|y - x| \quad \text{whenever } |y - x| < \tau \text{ and } q \text{ is a subgradient of } F \text{ at } y.$$

Let $A = \frac{1}{2}(A_0 + A_0^T)$. Then A is symmetric and

$$F(y) - F(x) - \langle a, y - x \rangle - \frac{1}{2} \langle A(y - x), y - x \rangle = o(|y - x|^2) \quad \text{as } y \rightarrow x,$$

i.e. for every $\kappa > 0$ the absolute value of the left-hand side is at most $\kappa|y - x|^2$ for all y in a neighbourhood of x .

Proof. $A^T = \frac{1}{2}(A_0^T + A_0) = A$, so A is symmetric. For every h , $\langle A_0^T h, h \rangle = \langle h, A_0 h \rangle$, hence $\langle Ah, h \rangle = \langle A_0 h, h \rangle$; it therefore suffices to prove the expansion with A_0 in place of A .

Fix $\kappa > 0$ and let $\tau > 0$ be as in the hypothesis. Let y with $|y - x| < \tau$ and put $h = y - x$. Let $N \geq 1$, $t_k = k/N$ and $y_k = x + t_k h$ for $k = 0, \dots, N$; then $y_0 = x$, $y_N = y$ and $|y_k - x| = t_k|h| < \tau$. By `convex_exists_subgradient` choose a subgradient q_k of F at y_k for each k , and write $q_k = a + t_k A_0 h + r_k$; by hypothesis $|r_k| \leq \kappa t_k |h| \leq \kappa |h|$.

Two-sided increments. Since $y_{k+1} - y_k = h/N$, the subgradient inequality at y_k (evaluated at y_{k+1}) and at y_{k+1} (evaluated at y_k) give

$$\frac{1}{N} \langle q_k, h \rangle \leq F(y_{k+1}) - F(y_k) \leq \frac{1}{N} \langle q_{k+1}, h \rangle \quad (0 \leq k \leq N-1).$$

Summing over k ,

$$\frac{1}{N} \sum_{k=0}^{N-1} \langle q_k, h \rangle \leq F(y) - F(x) \leq \frac{1}{N} \sum_{k=1}^N \langle q_k, h \rangle.$$

Evaluation of the sums. $\langle q_k, h \rangle = \langle a, h \rangle + t_k \langle A_0 h, h \rangle + \langle r_k, h \rangle$ with $|\langle r_k, h \rangle| \leq \kappa |h|^2$; also $\frac{1}{N} \sum_{k=0}^{N-1} t_k = \frac{N-1}{2N}$ and $\frac{1}{N} \sum_{k=1}^N t_k = \frac{N+1}{2N}$. Hence

$$\langle a, h \rangle + \frac{N-1}{2N} \langle A_0 h, h \rangle - \kappa |h|^2 \leq F(y) - F(x) \leq \langle a, h \rangle + \frac{N+1}{2N} \langle A_0 h, h \rangle + \kappa |h|^2.$$

Letting $N \rightarrow \infty$,

$$|F(y) - F(x) - \langle a, h \rangle - \frac{1}{2} \langle A_0 h, h \rangle| \leq \kappa |h|^2.$$

This holds for all $y \in B(x, \tau)$, a neighbourhood of x . As $\kappa > 0$ was arbitrary and $\langle A_0 h, h \rangle = \langle Ah, h \rangle$, the claimed $o(|y - x|^2)$ expansion follows. $\square$

2.5. lipschitz_image_null.

Lemma 2.5 (Lipschitz images of null sets are null). *Let $n \geq 0$, let vol be Lebesgue measure on $\mathbb{R}^n$ (Lean: `volume on EuclideanSpace  $\mathbb{R}$  (Fin  $n$ )`), let $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$, $s \subseteq \mathbb{R}^n$ and $K \geq 0$, and assume $|f(x) - f(x')| \leq K|x - x'|$ for all $x, x' \in s$. If $\text{vol}(s) = 0$ (outer measure), then $\text{vol}(f(s)) = 0$.*

Proof. If $n = 0$, $\mathbb{R}^0$ is a single point of measure 1, so $\text{vol}(s) = 0$ forces $s = \emptyset$ and $f(s) = \emptyset$. Let $n \geq 1$. (Lebesgue measure on Euclidean $\mathbb{R}^n$ is the product Lebesgue measure in the standard coordinates, the coordinate map being a measure-preserving isometry; closed cubes below are axis-parallel in these coordinates.)

Let $\eta > 0$. By outer regularity of Lebesgue measure there is an open set $U \supseteq s$ with $\text{vol}(U) < \eta$. Every open subset of $\mathbb{R}^n$ is a countable union of closed dyadic cubes Q_i with pairwise disjoint interiors, so $\sum_i \text{vol}(Q_i) = \text{vol}(U) < \eta$. Let ℓ_i be the side of Q_i , so $\text{vol}(Q_i) = \ell_i^n$ and $\text{diam } Q_i = \sqrt{n} \ell_i$.

Let I be the set of i with $Q_i \cap s \neq \emptyset$, and for $i \in I$ fix $s_i \in Q_i \cap s$. For $x \in Q_i \cap s$, $|f(x) - f(s_i)| \leq K|x - s_i| \leq K\sqrt{n} \ell_i$, so $f(Q_i \cap s)$ lies in the closed ball of radius $K\sqrt{n} \ell_i$ about

$f(s_i)$, which lies in the closed cube centred at $f(s_i)$ of side $2K\sqrt{n}\ell_i$, of volume $(2K\sqrt{n})^n\ell_i^n$. Since $s = \bigcup_{i \in I} (Q_i \cap s)$, we have $f(s) = \bigcup_{i \in I} f(Q_i \cap s)$, and countable subadditivity gives

$$\text{vol}(f(s)) \leq \sum_{i \in I} (2K\sqrt{n})^n \ell_i^n \leq (2K\sqrt{n})^n \sum_i \text{vol}(Q_i) < (2K\sqrt{n})^n \eta.$$

As $\eta > 0$ is arbitrary, $\text{vol}(f(s)) = 0$. $\square$

2.6. alexandrov_convex.

Theorem 2.6 (Alexandrov's theorem for convex functions). *Let $n \geq 0$ and let $F : \mathbb{R}^n \rightarrow \mathbb{R}$ be convex on $\mathbb{R}^n$. Then for Lebesgue-almost every $x \in \mathbb{R}^n$ there exist $a \in \mathbb{R}^n$ and a symmetric real $n \times n$ matrix A such that*

$$F(y) - F(x) - \langle a, y - x \rangle - \frac{1}{2} \langle A(y - x), y - x \rangle = o(|y - x|^2) \quad \text{as } y \rightarrow x$$

(for every $\kappa > 0$ the absolute value of the left-hand side is at most $\kappa|y - x|^2$ for y in a neighbourhood of x).

Proof. Call q a subgradient of F at y , and write $q \in \partial F(y)$, if $F(y) + \langle q, w - y \rangle \leq F(w)$ for all w . By `convex_exists_subgradient`, $\partial F(y) \neq \emptyset$ for every y . Let $P : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be the map of `convex_prox_map`: (i) $z - P(z) \in \partial F(P(z))$ for all z ; (ii) P is 1-Lipschitz; (iii) $P(y + q) = y$ whenever $q \in \partial F(y)$. Let vol denote Lebesgue measure.

Step 1 (exceptional set). By Rademacher's theorem (a Lipschitz map $\mathbb{R}^n \rightarrow \mathbb{R}^n$ is differentiable at almost every point; Mathlib `LipschitzWith.ae_differentiableAt`), the set N_2 of points where P is not differentiable has $\text{vol}(N_2) = 0$. By `lipschitz_image_null` (with $s = N_2$, $K = 1$), $\text{vol}(P(N_2)) = 0$. Let S be the set of z at which P is differentiable with $\det DP(z) = 0$. By the null-image theorem for singular derivatives (the easy case of Sard's theorem: if f is differentiable at every point of a set S with a derivative of zero determinant there, then $\text{vol}(f(S)) = 0$; Mathlib `addHaar_image_eq_zero_of_det_fderivWithin_eq_zero`), $\text{vol}(P(S)) = 0$. Put $N = P(N_2) \cup P(S)$; then $\text{vol}(N) = 0$. We show that the conclusion holds at every $x \notin N$; this proves the theorem.

Step 2 (a good preimage). Fix $x \notin N$. Choose $a \in \partial F(x)$ and put $z = x + a$; by (iii), $P(z) = x$. If $z \in N_2$ then $x = P(z) \in P(N_2)$, and if $z \in S$ then $x \in P(S)$; both are excluded. Hence P is differentiable at z and $D := DP(z)$ is an invertible linear map. Let $B = D^{-1}$ and $m = 1/(\|B\| + 1) > 0$; then $|v| = |BDv| \leq \|B\| |Dv|$ gives $|Dv| \geq m|v|$, and $|Bv| \leq |v|/m$, for all v .

Step 3 ($\partial F(x) = \{a\}$). Let $a' \in \partial F(x)$. For $t \in [0, 1]$, $a_t = (1 - t)a + ta' \in \partial F(x)$ (a convex combination of the two subgradient inequalities), so by (iii) $P(z + t(a' - a)) = P(x + a_t) = x = P(z)$. Differentiability of P at z gives $P(z + t(a' - a)) - P(z) - tD(a' - a) = o(t)$ as $t \rightarrow 0^+$, i.e. $tD(a' - a) = o(t)$, so $D(a' - a) = 0$ and, D being injective, $a' = a$.

Step 4 (subgradients near x are close to a). Claim: for every $\sigma > 0$ there is $\tau > 0$ such that $|q - a| < \sigma$ whenever $|y - x| < \tau$ and $q \in \partial F(y)$. Suppose not: then there are $\sigma > 0$, $y_k \rightarrow x$ and $q_k \in \partial F(y_k)$ with $|q_k - a| \geq \sigma$. By `convex_locally_lipschitz` with centre x and radius 1 there is Λ with $|q| \leq \Lambda$ for $q \in \partial F(y)$, $|y - x| \leq 1$, and $|F(y) - F(x)| \leq \Lambda|y - x|$ for $|y - x| \leq 1$. For large k , $|y_k - x| \leq 1$, so $|q_k| \leq \Lambda$, and a subsequence converges to some q^* with $|q^* - a| \geq \sigma$. For each fixed w , $F(w) \geq F(y_k) + \langle q_k, w - y_k \rangle$; letting $k \rightarrow \infty$ along the subsequence ($F(y_k) \rightarrow F(x)$ by the Lipschitz bound) gives $F(w) \geq F(x) + \langle q^*, w - x \rangle$. So $q^* \in \partial F(x) = \{a\}$, contradicting $|q^* - a| \geq \sigma$.

Step 5 (first-order expansion of subgradients). Let $A'_0 = B - \text{Id}$ (a linear map of $\mathbb{R}^n$). Claim: for every $\kappa > 0$ there is $\tau > 0$ such that $|q - a - A'_0(y - x)| \leq \kappa|y - x|$ whenever $|y - x| < \tau$ and $q \in \partial F(y)$. Fix $\kappa > 0$ and put $\theta = \min(m/2, \kappa m^2/2) > 0$. By differentiability of P at z choose $\tau_\theta > 0$ with $|P(z') - P(z) - D(z' - z)| \leq \theta|z' - z|$ for $|z' - z| < \tau_\theta$. By Step 4 choose $\tau \in (0, \tau_\theta/2]$ such that $|q - a| < \tau_\theta/2$ whenever $|y - x| < \tau$, $q \in \partial F(y)$. Now let such y, q be given and put $z' = y + q$. By (iii), $P(z') = y$; and $z' - z = (y - x) + (q - a)$, so $|z' - z| < \tau_\theta$. Let $e = P(z') - P(z) - D(z' - z) = (y - x) - D(z' - z)$; then $|e| \leq \theta|z' - z|$. Hence

$$m|z' - z| \leq |D(z' - z)| = |(y - x) - e| \leq |y - x| + \frac{m}{2}|z' - z|, \quad \text{so} \quad |z' - z| \leq \frac{2}{m}|y - x|.$$

Applying B to $D(z' - z) = (y - x) - e$ gives $z' - z - B(y - x) = -Be$, so $|z' - z - B(y - x)| \leq |e|/m \leq \frac{\theta}{m}|z' - z| \leq \frac{2\theta}{m^2}|y - x| \leq \kappa|y - x|$. Finally $q - a - A'_0(y - x) = (z' - z) - (y - x) - B(y - x) + (y - x) = z' - z - B(y - x)$, which proves the claim.

Step 6 (conclusion). Let A_0 be the matrix of A'_0 in the standard basis, $(A_0)_{ij} = \langle e_i, A'_0 e_j \rangle$, so that $A_0 h = A'_0 h$ for all h . By Step 5 and `convex_second_order_of_subgradient_expansion` (applied to F , x , a , A_0), the symmetric matrix $A = \frac{1}{2}(A_0 + A_0^T)$ satisfies $F(y) - F(x) - \langle a, y - x \rangle - \frac{1}{2}\langle A(y - x), y - x \rangle = o(|y - x|^2)$ as $y \rightarrow x$. As this holds for every $x \notin N$ and $\text{vol}(N) = 0$, the theorem is proved. $\square$

2.7. `c2_local_max_to_strict_quadratic`.

Lemma 2.7 (Reduction to a strict maximum against a quadratic). *Let $n \geq 0$, let $\Omega \subseteq \mathbb{R}^n$ be open, let $\psi : \mathbb{R}^n \rightarrow \mathbb{R}$ be C^2 on Ω , let $w : \mathbb{R}^n \rightarrow \mathbb{R}$, and let $x_0 \in \Omega$ be such that $w - \psi$ attains a local maximum at x_0 . Let $\eta > 0$. Write $H = \nabla^2 \psi(x_0)$ for the Hessian matrix of `EuclidHessian`, I for the identity matrix, and*

$$Q_\eta(y) = \psi(x_0) + \langle \nabla \psi(x_0), y - x_0 \rangle + \frac{1}{2} \langle (H + 2\eta I)(y - x_0), y - x_0 \rangle.$$

Then H is symmetric, and there is $r > 0$ such that $\overline{B}(x_0, r) \subseteq \Omega$ and

$$w(y) - Q_\eta(y) \leq w(x_0) - \psi(x_0) - \frac{\eta}{2}|y - x_0|^2 \quad \text{for all } y \in \overline{B}(x_0, r).$$

In particular (as $Q_\eta(x_0) = \psi(x_0)$) $w - Q_\eta$ has a strict maximum over $\overline{B}(x_0, r)$ at x_0 .

Proof. Step 1 (regularity). Since Ω is open and ψ is C^2 on Ω , ψ is differentiable at every point of Ω , the map $y \mapsto D\psi(y)$ is C^1 on Ω , and $D^2\psi = D(D\psi)$ exists at every point of Ω and is continuous on Ω as a map into bilinear forms (operator norm $\|\cdot\|$).

Step 2 (symmetry of H). ψ is C^2 on the neighbourhood Ω of x_0 , so by Schwarz's theorem $D^2\psi(x_0)(h)(k) = D^2\psi(x_0)(k)(h)$ for all h, k (Mathlib: `ContDiffAt.isSymmSndFderivAt`). Hence $H_{ij} = D^2\psi(x_0)(e_i)(e_j) = D^2\psi(x_0)(e_j)(e_i) = H_{ji}$.

Step 3 (identities). For $h = \sum_i h_i e_i$, bilinearity gives $D^2\psi(x_0)(h)(h) = \sum_{i,j} h_i h_j H_{ij} = \langle Hh, h \rangle$. By definition of the gradient, $D\psi(x_0)h = \langle \nabla \psi(x_0), h \rangle$. Also $\langle (H + 2\eta I)h, h \rangle = \langle Hh, h \rangle + 2\eta|h|^2$, so

$$(1) \quad Q_\eta(x_0 + h) = \psi(x_0) + D\psi(x_0)h + \frac{1}{2}D^2\psi(x_0)(h)(h) + \eta|h|^2.$$

Step 4 (choice of r). Choose $\rho > 0$ with $B(x_0, \rho) \subseteq \Omega$ (Ω open); choose $\rho' > 0$ with $w(y) - \psi(y) \leq w(x_0) - \psi(x_0)$ for $|y - x_0| < \rho'$ (local maximum); by continuity of $D^2\psi$ at x_0 (Step 1) choose $\rho'' \in (0, \rho]$ with $\|D^2\psi(y) - D^2\psi(x_0)\| \leq \eta$ for $|y - x_0| < \rho''$. Put $r = \frac{1}{2} \min(\rho, \rho', \rho'') > 0$. Then $\overline{B}(x_0, r) \subseteq B(x_0, \rho) \subseteq \Omega$.

Step 5 (Taylor). Let $y \in \overline{B}(x_0, r)$ and $h = y - x_0$, so $|h| \leq r$. For $|t| < 2$ we have $|th| < 2r \leq \rho$, so $x_0 + th \in \Omega$; thus $g(t) = \psi(x_0 + th)$ is C^2 on $(-2, 2)$ with $g'(t) = D\psi(x_0 + th)h$ and $g''(t) = D^2\psi(x_0 + th)(h)(h)$ by the chain rule. Taylor's theorem with Lagrange remainder on $[0, 1]$ gives $\theta \in (0, 1)$ with $g(1) = g(0) + g'(0) + \frac{1}{2}g''(\theta)$ (if $h = 0$ this is trivial), i.e.

$$\psi(y) = \psi(x_0) + D\psi(x_0)h + \frac{1}{2}D^2\psi(x_0)(h)(h) + \frac{1}{2}(D^2\psi(x_0 + \theta h) - D^2\psi(x_0))(h)(h).$$

As $|\theta h| \leq r < \rho''$, the last term is at most $\frac{1}{2}\eta|h|^2$. Comparing with (1),

$$\psi(y) \leq Q_\eta(y) - \eta|h|^2 + \frac{1}{2}\eta|h|^2 = Q_\eta(y) - \frac{\eta}{2}|h|^2.$$

Step 6 (conclusion). Since $|h| \leq r < \rho'$, $w(y) - \psi(y) \leq w(x_0) - \psi(x_0)$. Therefore

$$w(y) - Q_\eta(y) \leq w(y) - \psi(y) - \frac{\eta}{2}|y - x_0|^2 \leq w(x_0) - \psi(x_0) - \frac{\eta}{2}|y - x_0|^2.$$

Finally $Q_\eta(x_0) = \psi(x_0)$, so the right-hand side equals $(w - Q_\eta)(x_0) - \frac{\eta}{2}|y - x_0|^2$, which is $< (w - Q_\eta)(x_0)$ for $y \neq x_0$. $\square$

2.8. disjointSum_continuousOn_locallyBddAbove.

Lemma 2.8 (Continuity and local upper bound of the disjoint sum). *Let $(\mathbf{X}, d_{\mathbf{X}})$, $(\mathbf{Y}, d_{\mathbf{Y}})$ be metric spaces and equip $\mathbf{X} \times \mathbf{Y}$ with the ℓ^1 metric d_1 . Let $\mathcal{U} \subseteq \mathbf{X}$, $\mathcal{V} \subseteq \mathbf{Y}$, $\mathcal{W} = \mathcal{U} \times \mathcal{V}$, and let $u : \mathbf{X} \rightarrow \mathbb{R}$, $v : \mathbf{Y} \rightarrow \mathbb{R}$ be such that u is continuous on $\mathcal{U}$ and v is continuous on $\mathcal{V}$ (as functions restricted to these sets). Assume u is locally bounded from above in $\mathcal{U}$ (every $x \in \mathcal{U}$ has a neighbourhood s in $\mathbf{X}$ such that u is bounded above on $s \cap \mathcal{U}$) and v is locally bounded from above in $\mathcal{V}$ (likewise). Then the disjoint sum $w(x, y) = u(x) + v(y)$ is continuous on $\mathcal{W}$, and w is locally bounded from above in $\mathcal{W}$: every $z \in \mathcal{W}$ has a neighbourhood t in $\mathbf{X} \times \mathbf{Y}$ such that w is bounded above on $t \cap \mathcal{W}$.*

Proof. The projections $\pi_1(x, y) = x$ and $\pi_2(x, y) = y$ are continuous on $(\mathbf{X} \times \mathbf{Y}, d_1)$ (they are 1-Lipschitz: $d_{\mathbf{X}}(x_1, x_2) \leq d_1((x_1, y_1), (x_2, y_2))$, and likewise for π_2).

Continuity. π_1 maps $\mathcal{W}$ into $\mathcal{U}$ and u is continuous on $\mathcal{U}$, so $u \circ \pi_1$ is continuous on $\mathcal{W}$; similarly $v \circ \pi_2$ is continuous on $\mathcal{W}$ since $\pi_2(\mathcal{W}) \subseteq \mathcal{V}$. Their sum w is continuous on $\mathcal{W}$.

Local upper bound. Let $z = (x, y) \in \mathcal{W}$, so $x \in \mathcal{U}$, $y \in \mathcal{V}$. Choose a neighbourhood s of x and $A \in \mathbb{R}$ with $u \leq A$ on $s \cap \mathcal{U}$, and a neighbourhood r of y and $B \in \mathbb{R}$ with $v \leq B$ on $r \cap \mathcal{V}$. Let $t = \pi_1^{-1}(s) \cap \pi_2^{-1}(r)$; by continuity of π_1, π_2 at z this is a neighbourhood of z . If $p = (p_1, p_2) \in t \cap \mathcal{W}$ then $p_1 \in s \cap \mathcal{U}$ and $p_2 \in r \cap \mathcal{V}$, whence $w(p) = u(p_1) + v(p_2) \leq A + B$. So w is bounded above by $A + B$ on $t \cap \mathcal{W}$. $\square$

2.9. euclidSplit_norm_sq.

Lemma 2.9 (Pythagoras for the splitting $\mathbb{R}^{n+m} = \mathbb{R}^n \times \mathbb{R}^m$). *Let $n, m \geq 0$ and $z \in \mathbb{R}^{n+m}$. With the blocks $\text{fst } z \in \mathbb{R}^n$ of **EuclidFst** and $\text{snd } z \in \mathbb{R}^m$ of **EuclidSnd**,*

$$|z|^2 = |\text{fst } z|^2 + |\text{snd } z|^2.$$

Proof. By definition of the Euclidean norm, $|z|^2 = \sum_{k=1}^{n+m} z_k^2$. Splitting the index set $\{1, \dots, n+m\}$ into $\{1, \dots, n\}$ and $\{n+1, \dots, n+m\}$ (Lean: `Fin.sum_univ_add`, indices `Fin.castAdd m i` and `Fin.natAdd n j`),

$$|z|^2 = \sum_{i=1}^n z_i^2 + \sum_{j=1}^m z_{n+j}^2 = \sum_{i=1}^n (\text{fst } z)_i^2 + \sum_{j=1}^m (\text{snd } z)_j^2 = |\text{fst } z|^2 + |\text{snd } z|^2.$$

$\square$

2.10. supConvolution_properties.

Lemma 2.10 (Basic properties of the sup-convolution). *Let $n \geq 0$, let $K \subseteq \mathbb{R}^n$ be compact and nonempty, let $u : \mathbb{R}^n \rightarrow \mathbb{R}$ be upper semicontinuous on K (relative to K), let $\varepsilon > 0$, and let $u^\varepsilon = u_K^\varepsilon$ be the sup-convolution of **SupConvolution**. Then:*

- (i) *for all $x \in \mathbb{R}^n$ and $y \in K$: $u(y) - \frac{|x-y|^2}{2\varepsilon} \leq u^\varepsilon(x)$;*
- (ii) *for every $x \in \mathbb{R}^n$ there is $\hat{x} \in K$ with $u^\varepsilon(x) = u(\hat{x}) - \frac{|x-\hat{x}|^2}{2\varepsilon}$;*
- (iii) *the function $x \mapsto u^\varepsilon(x) + \frac{|x|^2}{2\varepsilon}$ is convex on $\mathbb{R}^n$.*

Proof. Fix $x \in \mathbb{R}^n$ and let $\varphi_x(y) = u(y) - |x-y|^2/(2\varepsilon)$ and $S_x = \varphi_x(K)$, so that $u^\varepsilon(x) = \sup S_x$ by **SupConvolution**.

φ_x attains its maximum on K . φ_x is upper semicontinuous on K , being the sum of u (upper semicontinuous on K) and the continuous function $y \mapsto -|x-y|^2/(2\varepsilon)$. Let $s = \sup_K \varphi_x \in (-\infty, +\infty]$ (extended-real supremum; $K \neq \emptyset$) and choose $y_k \in K$ with $\varphi_x(y_k) \rightarrow s$. By compactness a subsequence converges to some $\hat{x} \in K$; along it, upper semicontinuity at $\hat{x}$ gives $\varphi_x(\hat{x}) \geq \limsup \varphi_x(y_k) = s$. Hence $s = \varphi_x(\hat{x})$ is finite and attained.

(i) and (ii). Thus S_x is nonempty, bounded above, and has the greatest element $\varphi_x(\hat{x})$. So its real supremum is $\sup S_x = \varphi_x(\hat{x})$ (Lean's convention for empty or unbounded sets does not apply), which is (ii), and every element of S_x is $\leq \sup S_x$, which is (i).

(iii). For $y \in K$ define the affine function $\ell_y(x) = u(y) - \frac{|y|^2}{2\varepsilon} + \frac{\langle x, y \rangle}{\varepsilon}$. Since $|x - y|^2 = |x|^2 - 2\langle x, y \rangle + |y|^2$, we have $\varphi_x(y) + \frac{|x|^2}{2\varepsilon} = \ell_y(x)$. Put $G(x) = u^\varepsilon(x) + \frac{|x|^2}{2\varepsilon}$. By (i), $\ell_y(x) \leq G(x)$ for all $x \in \mathbb{R}^n$, $y \in K$; by (ii), $G(x) = \ell_{\hat{x}}(x)$ for a suitable $\hat{x} \in K$. Now let $x_1, x_2 \in \mathbb{R}^n$, $t \in [0, 1]$, $x_t = tx_1 + (1-t)x_2$, and choose $\hat{x} \in K$ with $G(x_t) = \ell_{\hat{x}}(x_t)$. Since $\ell_{\hat{x}}$ is affine,

$$G(x_t) = t\ell_{\hat{x}}(x_1) + (1-t)\ell_{\hat{x}}(x_2) \leq tG(x_1) + (1-t)G(x_2).$$

As $\mathbb{R}^n$ is convex, G is convex on $\mathbb{R}^n$. $\square$

2.11. disjoint_supConvolution_concentration.

Lemma 2.11 (Concentration of the sup-convolutions in disjoint variables). *Let $n, m \geq 0$ and write $\text{fst} : \mathbb{R}^{n+m} \rightarrow \mathbb{R}^n$, $\text{snd} : \mathbb{R}^{n+m} \rightarrow \mathbb{R}^m$ for the block maps of [EuclidFst](#) and [EuclidSnd](#). Let $z_0 \in \mathbb{R}^{n+m}$, $r > 0$, $x_0 = \text{fst } z_0$, $y_0 = \text{snd } z_0$, $K_1 = \overline{B}(x_0, r) \subseteq \mathbb{R}^n$, $K_2 = \overline{B}(y_0, r) \subseteq \mathbb{R}^m$. Let $u : \mathbb{R}^n \rightarrow \mathbb{R}$ be upper semicontinuous on K_1 , $v : \mathbb{R}^m \rightarrow \mathbb{R}$ upper semicontinuous on K_2 (relative to these sets), and $Q : \mathbb{R}^{n+m} \rightarrow \mathbb{R}$ continuous. Assume*

$$u(\text{fst } z) + v(\text{snd } z) - Q(z) < m_0 := u(x_0) + v(y_0) - Q(z_0) \quad \text{for all } z \in \overline{B}(z_0, r) \setminus \{z_0\}.$$

For $\varepsilon > 0$ let $u^\varepsilon = u_{K_1}^\varepsilon$ and $v^\varepsilon = v_{K_2}^\varepsilon$ be the sup-convolutions of [SupConvolution](#). Then for every $\rho > 0$ there are $\beta > 0$ and $\varepsilon_0 > 0$ such that

$$u^\varepsilon(\text{fst } z) + v^\varepsilon(\text{snd } z) - Q(z) \leq m_0 - \beta$$

for all $\varepsilon \in (0, \varepsilon_0)$ and all $z \in \overline{B}(z_0, r)$ with $|z - z_0| \geq \rho$.

Proof. Preliminaries. fst, snd are linear (they select coordinates), and by [euclidSplit_norm_sq](#), $|h|^2 = |\text{fst } h|^2 + |\text{snd } h|^2$ for all h ; in particular $|\text{fst } h| \leq |h|$ and $|\text{snd } h| \leq |h|$, so both maps are 1-Lipschitz, hence continuous, and they map $B := \overline{B}(z_0, r)$ into K_1 and K_2 respectively. For $x \in \mathbb{R}^n$, $y \in \mathbb{R}^m$ let $x \oplus y \in \mathbb{R}^{n+m}$ denote the concatenation (first block x , second block y); by [euclidSplit_norm_sq](#), $|x \oplus y - z|^2 = |x - \text{fst } z|^2 + |y - \text{snd } z|^2$.

K_1, K_2 are compact and nonempty; u and v , being upper semicontinuous there, attain their maxima $M_u = \max_{K_1} u$, $M_v = \max_{K_2} v$ (a maximizing sequence has a convergent subsequence and upper semicontinuity at its limit bounds the limsup). Let $M_Q \geq \sup_B |Q|$ (Q continuous, B compact). By [supConvolution_properties](#)(ii) (applied to u on K_1 and v on K_2), for every $\varepsilon > 0$ and z there are $\hat{x} \in K_1$, $\hat{y} \in K_2$ with

$$(1) \quad u^\varepsilon(\text{fst } z) = u(\hat{x}) - \frac{|\text{fst } z - \hat{x}|^2}{2\varepsilon}, \quad v^\varepsilon(\text{snd } z) = v(\hat{y}) - \frac{|\text{snd } z - \hat{y}|^2}{2\varepsilon}.$$

The gap β . Fix $\rho > 0$ and let $A_\rho = \{z \in B : |z - z_0| \geq \rho\}$, a compact set. If $A_\rho = \emptyset$ the claim holds vacuously with $\beta = \varepsilon_0 = 1$. Otherwise $\Phi(z) = u(\text{fst } z) + v(\text{snd } z) - Q(z)$ is upper semicontinuous on A_ρ : at $z^* \in A_\rho$ and for $t > 0$, upper semicontinuity of u on K_1 at $\text{fst } z^*$ together with continuity of fst and $\text{fst}(A_\rho) \subseteq K_1$ gives $u(\text{fst } z) < u(\text{fst } z^*) + t$ for $z \in A_\rho$ near z^* , and similarly for v and $-Q$. Hence Φ attains its maximum m_ρ on A_ρ at some z_ρ ; as $|z_\rho - z_0| \geq \rho > 0$, $z_\rho \neq z_0$, so $m_\rho < m_0$ by hypothesis. Put $\beta = \frac{1}{2}(m_0 - m_\rho) > 0$, so $m_\rho + \beta = m_0 - \beta$.

Local control. For each $z^* \in A_\rho$ choose $\eta(z^*) > 0$ such that: $u(x) < u(\text{fst } z^*) + \beta/3$ for $x \in K_1$ with $|x - \text{fst } z^*| < \eta(z^*)$; $v(y) < v(\text{snd } z^*) + \beta/3$ for $y \in K_2$ with $|y - \text{snd } z^*| < \eta(z^*)$ (upper semicontinuity of u on K_1 at $\text{fst } z^* \in K_1$ and of v on K_2 at $\text{snd } z^* \in K_2$); and $Q(z) > Q(z^*) - \beta/3$ for $|z - z^*| < \eta(z^*)$ (continuity of Q). The open balls $B(z^*, \eta(z^*))$, $z^* \in A_\rho$, cover the compact set A_ρ ; by the Lebesgue number lemma there is $\lambda > 0$ such that for every $z \in A_\rho$ there is $z^* \in A_\rho$ with $B(z, \lambda) \subseteq B(z^*, \eta(z^*))$.

Choice of ε_0 and conclusion. Let $C_0 = M_u + M_v + M_Q - m_0 + \beta$ and $\varepsilon_0 = \lambda^2/(2(|C_0| + 1)) > 0$. Let $0 < \varepsilon < \varepsilon_0$ and $z \in A_\rho$, and suppose for contradiction that $W := u^\varepsilon(\text{fst } z) + v^\varepsilon(\text{snd } z) - Q(z) > m_0 - \beta$. With $\hat{x}, \hat{y}$ as in (1),

$$\frac{|\text{fst } z - \hat{x}|^2 + |\text{snd } z - \hat{y}|^2}{2\varepsilon} = u(\hat{x}) + v(\hat{y}) - Q(z) - W < M_u + M_v + M_Q - m_0 + \beta = C_0 \leq |C_0|,$$

using $u(\hat{x}) \leq M_u$, $v(\hat{y}) \leq M_v$, $-Q(z) \leq M_Q$. Hence the point $\zeta = \hat{x} \oplus \hat{y}$ satisfies $|\zeta - z|^2 = |\hat{x} - \text{fst } z|^2 + |\hat{y} - \text{snd } z|^2 < 2\varepsilon|C_0| < \lambda^2$, i.e. $\zeta \in B(z, \lambda)$. Choose $z^* \in A_\rho$ with $B(z, \lambda) \subseteq B(z^*, \eta :=$

$\eta(z^*)$). Then $|z - z^*| < \eta$, and $|\hat{x} - \text{fst } z^*| = |\text{fst}(\zeta - z^*)| \leq |\zeta - z^*| < \eta$, $|\hat{y} - \text{snd } z^*| \leq |\zeta - z^*| < \eta$ (as $\text{fst } \zeta = \hat{x}$, $\text{snd } \zeta = \hat{y}$ and the block maps are linear and 1-Lipschitz). Since $\hat{x} \in K_1$, $\hat{y} \in K_2$, the local control gives

$$u(\hat{x}) + v(\hat{y}) - Q(z) < u(\text{fst } z^*) + v(\text{snd } z^*) - Q(z^*) + \beta = \Phi(z^*) + \beta \leq m_\rho + \beta = m_0 - \beta.$$

But by (1), $W \leq u(\hat{x}) + v(\hat{y}) - Q(z)$ (the subtracted squares are nonnegative), so $W < m_0 - \beta$, a contradiction. Hence $W \leq m_0 - \beta$ for all $\varepsilon \in (0, \varepsilon_0)$ and all $z \in B$ with $|z - z_0| \geq \rho$. $\square$

2.12. euclidSplit_preimage_null.

Lemma 2.12 (Preimages of null sets under the block projections are null). *Let $n, m \geq 0$ and let vol denote Lebesgue (outer) measure on $\mathbb{R}^n$, $\mathbb{R}^m$ and $\mathbb{R}^{n+m}$ (Lean: `volume` on the corresponding `EuclideanSpace`). With the block maps of `EuclidFst` and `EuclidSnd`:*

- (i) *if $N \subseteq \mathbb{R}^n$ has $\text{vol}(N) = 0$, then $\text{vol}\{z \in \mathbb{R}^{n+m} : \text{fst } z \in N\} = 0$;*
- (ii) *if $N \subseteq \mathbb{R}^m$ has $\text{vol}(N) = 0$, then $\text{vol}\{z \in \mathbb{R}^{n+m} : \text{snd } z \in N\} = 0$.*

Proof. (i) Since N is null, there is a Borel set $N' \supseteq N$ with $\text{vol}(N') = 0$ (a measurable hull). Under the concatenation of coordinates $\mathbb{R}^{n+m} \cong \mathbb{R}^n \times \mathbb{R}^m$, $z \mapsto (\text{fst } z, \text{snd } z)$ (a bijection that merely relabels coordinates), Lebesgue measure on $\mathbb{R}^{n+m}$ corresponds to the product of Lebesgue measures on $\mathbb{R}^n$ and $\mathbb{R}^m$ (both are the product of $n + m$ copies of one-dimensional Lebesgue measure, and the Euclidean-space volume is the coordinate product measure). The set $\{z : \text{fst } z \in N'\}$ corresponds to $N' \times \mathbb{R}^m = \bigcup_{k \geq 1} N' \times [-k, k]^m$, and each $N' \times [-k, k]^m$ is a Borel set of product measure $\text{vol}(N') (2k)^m = 0$. By countable subadditivity $\text{vol}\{z : \text{fst } z \in N'\} = 0$, and by monotonicity $\text{vol}\{z : \text{fst } z \in N\} = 0$.

(ii) Identical, with $\mathbb{R}^n \times N' = \bigcup_k [-k, k]^n \times N'$.

(Equivalently: `fst` and `snd` are surjective linear maps, and the preimage of a Lebesgue-null set under a surjective linear map between Euclidean spaces is Lebesgue-null; `Mathlib ae_comp_linearMap_mem_iff`.) $\square$

2.13. expansion_sub_half_sq_norm.

Lemma 2.13 (Second-order expansion after removing $|x|^2/(2\varepsilon)$). *Let $n \geq 0$, $w : \mathbb{R}^n \rightarrow \mathbb{R}$, $\varepsilon > 0$, $y, a_0 \in \mathbb{R}^n$ and let A_0 be a real $n \times n$ matrix. Put $F(z) = w(z) + \frac{|z|^2}{2\varepsilon}$ and assume*

$$F(z) - F(y) - \langle a_0, z - y \rangle - \frac{1}{2} \langle A_0(z - y), z - y \rangle = o(|z - y|^2) \quad (z \rightarrow y).$$

Let $a = a_0 - \frac{1}{\varepsilon}y$ and $A = A_0 - \frac{1}{\varepsilon}I$. Then for every $\kappa > 0$, for all z in a neighbourhood of y ,

$$|w(z) - w(y) - \langle a, z - y \rangle - \frac{1}{2} \langle A(z - y), z - y \rangle| \leq \kappa |z - y|^2.$$

Proof. Let $h = z - y$. Since $|z|^2 = |y|^2 + 2\langle y, h \rangle + |h|^2$, $\langle a, h \rangle = \langle a_0, h \rangle - \frac{1}{\varepsilon} \langle y, h \rangle$ and $\langle Ah, h \rangle = \langle A_0h, h \rangle - \frac{1}{\varepsilon} |h|^2$,

$$w(z) - w(y) - \langle a, h \rangle - \frac{1}{2} \langle Ah, h \rangle = F(z) - F(y) - \frac{2\langle y, h \rangle + |h|^2}{2\varepsilon} - \langle a_0, h \rangle + \frac{\langle y, h \rangle}{\varepsilon} - \frac{1}{2} \langle A_0h, h \rangle + \frac{|h|^2}{2\varepsilon} = F(z) - F(y) - \langle a_0, h \rangle$$

By the little- o hypothesis, for every $\kappa > 0$ the absolute value of the right-hand side is at most $\kappa |h|^2$ for z near y , which is the claim. $\square$

2.14. jensen_positive_measure.

Lemma 2.14 (Jensen's lemma). *Let $n \geq 0$, $W : \mathbb{R}^n \rightarrow \mathbb{R}$ and $C \geq 0$ be such that $\Phi(x) = W(x) + \frac{C}{2}|x|^2$ is convex on $\mathbb{R}^n$. Let $x_0, x_* \in \mathbb{R}^n$, $r > 0$, $\delta > 0$ with $x_* \in \overline{B}(x_0, r)$, and assume*

$$W(y) + 2\delta r < W(x_*) \quad \text{for all } y \text{ with } |y - x_0| = r.$$

Let K_δ be the set of $x \in B(x_0, r)$ for which there is $p \in \mathbb{R}^n$ with $|p| \leq \delta$ and

$$W(y) + \langle p, y \rangle \leq W(x) + \langle p, x \rangle \quad \text{for all } y \in \overline{B}(x_0, r).$$

Then K_δ has positive Lebesgue measure: $\text{vol}(K_\delta) > 0$.

Proof. Step 1 (W is continuous). For every y_0 , `convex_locally_lipschitz`(i) applied to the convex Φ with centre y_0 and radius 1 gives $|\Phi(y) - \Phi(y_0)| \leq \Lambda|y - y_0|$ for $|y - y_0| \leq 1$; so Φ is continuous, and so is $W = \Phi - \frac{C}{2}|\cdot|^2$.

Step 2 (every small p is attained in K_δ). Let $|p| \leq \delta$. The continuous function $W + \langle p, \cdot \rangle$ attains its maximum over the compact nonempty ball $\overline{B}(x_0, r)$ at some x_p . If $|x_p - x_0| = r$, then using $|\langle p, v \rangle| \leq \delta|v|$, the hypothesis, and $|x_* - x_0| \leq r$,

$$W(x_p) + \langle p, x_p \rangle \leq W(x_p) + \delta r + \langle p, x_0 \rangle < W(x_*) - \delta r + \langle p, x_0 \rangle \leq W(x_*) + \langle p, x_* \rangle,$$

contradicting maximality (as $x_* \in \overline{B}(x_0, r)$). Hence $x_p \in B(x_0, r)$ and $x_p \in K_\delta$ with witness p .

Step 3 (two-sided bounds and uniqueness of p). Let $x \in K_\delta$ with witness p . For $y \in \overline{B}(x_0, r)$, $W(y) \leq W(x) - \langle p, y - x \rangle$; adding $\frac{C}{2}|y|^2 = \frac{C}{2}|x|^2 + C\langle x, y - x \rangle + \frac{C}{2}|y - x|^2$,

$$(U) \quad \Phi(y) \leq \Phi(x) + \langle Cx - p, y - x \rangle + \frac{C}{2}|y - x|^2 \quad (y \in \overline{B}(x_0, r)).$$

By `convex_exists_subgradient` there is q with $\Phi(y) \geq \Phi(x) + \langle q, y - x \rangle$ for all y . Let e be a unit vector. As x lies in the open ball, $x \pm te \in \overline{B}(x_0, r)$ for small $t > 0$, and (U) with the lower bound gives $\pm t\langle q - (Cx - p), e \rangle \leq \frac{C}{2}t^2$; dividing by t and letting $t \rightarrow 0^+$ yields $\langle q - (Cx - p), e \rangle = 0$. Hence $q = Cx - p$, i.e. $p = Cx - q$. Since q was fixed before p , the witness p of x is unique; denote it $\pi(x)$, and put $s_x = Cx - \pi(x)$. Then

$$(L,U) \quad \Phi(y) \geq \Phi(x) + \langle s_x, y - x \rangle \quad (y \in \mathbb{R}^n), \quad \Phi(y) \leq \Phi(x) + \langle s_x, y - x \rangle + \frac{C}{2}|y - x|^2 \quad (y \in \overline{B}(x_0, r)).$$

By Step 2 and uniqueness, $\pi(x_p) = p$ for every $|p| \leq \delta$, so $\overline{B}(0, \delta) \subseteq \pi(K_\delta)$.

Step 4 (local Lipschitz bound). Let $\theta > 0$ and $x_1, x_2 \in K_\delta$ with $|x_i - x_0| \leq r - \theta$ and $d = |x_1 - x_2| \leq \theta/2$. Write $s_i = s_{x_i}$. We show $|s_1 - s_2| \leq 2Cd$. If $d = 0$ or $s_1 = s_2$ this is clear; otherwise let $e = (s_2 - s_1)/|s_2 - s_1|$, $t = 2d \leq \theta$ and $y = x_1 + te \in \overline{B}(x_0, r)$. By (L) at x_2 and (U) at x_1 : $\Phi(x_2) + \langle s_2, y - x_2 \rangle \leq \Phi(y) \leq \Phi(x_1) + \langle s_1, y - x_1 \rangle + \frac{C}{2}t^2$. By (L) at x_1 evaluated at x_2 : $\Phi(x_2) \geq \Phi(x_1) + \langle s_1, x_2 - x_1 \rangle$. Combining, $\langle s_2 - s_1, y - x_2 \rangle \leq \frac{C}{2}t^2$. Since $y - x_2 = (x_1 - x_2) + te$,

$$t|s_2 - s_1| = \langle s_2 - s_1, te \rangle \leq \frac{C}{2}t^2 + |s_2 - s_1|d, \quad \text{so} \quad (t - d)|s_2 - s_1| \leq \frac{C}{2}t^2,$$

and with $t = 2d$ this is $|s_2 - s_1| \leq 2Cd$. Consequently $|\pi(x_1) - \pi(x_2)| = |C(x_1 - x_2) - (s_1 - s_2)| \leq 3C|x_1 - x_2|$.

Step 5 (conclusion). Define $g : \mathbb{R}^n \rightarrow \mathbb{R}^n$ by $g = \pi$ on K_δ and $g = 0$ elsewhere. Fix an enumeration $(c_j)_{j \geq 1}$ of the points of $\mathbb{R}^n$ with rational coordinates. For integers $m, j \geq 1$ let

$$T_{m,j} = K_\delta \cap \overline{B}(x_0, r - \frac{1}{m}) \cap \overline{B}(c_j, \frac{1}{4m}).$$

Any two points of $T_{m,j}$ are at distance $\leq \frac{1}{2m}$, so Step 4 with $\theta = \frac{1}{m}$ shows $|g(x_1) - g(x_2)| \leq 3C|x_1 - x_2|$ for $x_1, x_2 \in T_{m,j}$. Every $x \in K_\delta$ satisfies $|x - x_0| < r$, hence $|x - x_0| \leq r - \frac{1}{m}$ for some m , and $x \in \overline{B}(c_j, \frac{1}{4m})$ for some j by density of rational points; so $K_\delta = \bigcup_{m,j} T_{m,j}$ and $\pi(K_\delta) = \bigcup_{m,j} g(T_{m,j})$.

Suppose $\text{vol}(K_\delta) = 0$. Then $\text{vol}(T_{m,j}) = 0$, so by `lipschitz_image_null` (with $f = g$, $s = T_{m,j}$, constant $3C$) $\text{vol}(g(T_{m,j})) = 0$ for all m, j , and by countable subadditivity $\text{vol}(\pi(K_\delta)) = 0$. But $\pi(K_\delta) \supseteq \overline{B}(0, \delta)$ by Step 3, and $\text{vol}(\overline{B}(0, \delta)) > 0$ since $\delta > 0$. This contradiction shows $\text{vol}(K_\delta) > 0$. $\square$

2.15. jet_twosided_to_upper.

Lemma 2.15 (Upper half of a two-sided second-order expansion). *Let $k \geq 0$, $w : \mathbb{R}^k \rightarrow \mathbb{R}$, $y, a_0 \in \mathbb{R}^k$ and let A_0 be a real $k \times k$ matrix. Assume that for every $\kappa > 0$, for all x in a neighbourhood of y ,*

$$|w(x) - w(y) - \langle a_0, x - y \rangle - \frac{1}{2}\langle A_0(x - y), x - y \rangle| \leq \kappa|x - y|^2.$$

Then for every $\kappa > 0$, for all x in a neighbourhood of y ,

$$w(x) \leq w(y) + \langle a_0, x - y \rangle + \frac{1}{2}\langle A_0(x - y), x - y \rangle + \kappa|x - y|^2.$$

Proof. Fix $\kappa > 0$ and take the neighbourhood of y given by the hypothesis for this κ . For x in it, $|t| \leq s$ implies $t \leq s$, so with $t = w(x) - w(y) - \langle a_0, x - y \rangle - \frac{1}{2} \langle A_0(x - y), x - y \rangle$ and $s = \kappa|x - y|^2$ we get $t \leq s$; adding $w(y) + \langle a_0, x - y \rangle + \frac{1}{2} \langle A_0(x - y), x - y \rangle$ to both sides gives the claimed inequality on the same neighbourhood. $\square$

2.16. quadratic_convex_part.

Lemma 2.16 (Convexity of the quadratic correction). *Let $n \geq 0$, $x_0, c \in \mathbb{R}^n$, $q_0, \lambda \in \mathbb{R}$, and let P be a symmetric real $n \times n$ matrix with $\langle Ph, h \rangle \leq \lambda|h|^2$ for all $h \in \mathbb{R}^n$. Put $Q(x) = q_0 + \langle c, x - x_0 \rangle + \frac{1}{2} \langle P(x - x_0), x - x_0 \rangle$. Then the function*

$$x \mapsto \frac{\lambda}{2}|x|^2 - Q(x)$$

is convex on $\mathbb{R}^n$.

Proof. Let $\kappa(x) = \frac{\lambda}{2}|x|^2 - Q(x)$. Fix $x, y \in \mathbb{R}^n$ and put $d = x - y$. For $t \in \mathbb{R}$, expanding $|y + td|^2 = |y|^2 + 2t\langle y, d \rangle + t^2|d|^2$ and, using the symmetry $\langle Pd, y - x_0 \rangle = \langle P(y - x_0), d \rangle$,

$$\langle P(y - x_0 + td), y - x_0 + td \rangle = \langle P(y - x_0), y - x_0 \rangle + 2t\langle P(y - x_0), d \rangle + t^2\langle Pd, d \rangle,$$

we get

$$\kappa(y + td) = \kappa(y) + tK_1 + t^2K_2, \quad K_1 = \lambda\langle y, d \rangle - \langle c, d \rangle - \langle P(y - x_0), d \rangle, \quad K_2 = \frac{\lambda}{2}|d|^2 - \frac{1}{2}\langle Pd, d \rangle.$$

By hypothesis $K_2 \geq 0$. Let $a, b \geq 0$ with $a + b = 1$. Then $ax + by = y + ad$ and $x = y + 1 \cdot d$, so $a\kappa(x) + b\kappa(y) - \kappa(ax + by) = a(\kappa(y) + K_1 + K_2) + (1 - a)\kappa(y) - (\kappa(y) + aK_1 + a^2K_2) = (a - a^2)K_2 \geq 0$, since $0 \leq a \leq 1$ gives $a^2 \leq a$. Hence κ is convex. $\square$

2.17. quadratic_upper_bound_conditions.

Lemma 2.17 (First- and second-order conditions from a local quadratic upper bound). *Let $n \geq 0$, $\gamma \in \mathbb{R}^n$ and let G be a symmetric real $n \times n$ matrix. Assume that for every $\kappa > 0$ there is $\eta > 0$ such that*

$$\langle \gamma, h \rangle + \frac{1}{2} \langle Gh, h \rangle \leq \kappa|h|^2 \quad \text{for all } h \in \mathbb{R}^n \text{ with } |h| < \eta.$$

Then $\gamma = 0$ and $-G$ is positive semidefinite, i.e. $\langle Ge, e \rangle \leq 0$ for all $e \in \mathbb{R}^n$.

Proof. Write $q(h) = \langle Gh, h \rangle$; then $q(th) = t^2q(h)$ for $t \in \mathbb{R}$.

$\gamma = 0$. Take $\kappa = 1$ and the corresponding η . Suppose $\gamma \neq 0$ and put $Y = 1 - \frac{q(\gamma)}{2|\gamma|^2}$ and $t = \min(\frac{\eta}{2|\gamma|}, \frac{1}{2(|Y|+1)}) > 0$. Then $|t\gamma| = t|\gamma| \leq \eta/2 < \eta$, so with $h = t\gamma$ the hypothesis gives $t|\gamma|^2 + \frac{1}{2}t^2q(\gamma) \leq t^2|\gamma|^2$. Dividing by $t|\gamma|^2 > 0$ gives $1 \leq tY \leq t|Y| \leq \frac{|Y|}{2(|Y|+1)} < 1$, a contradiction. Hence $\gamma = 0$.

$q \leq 0$. Fix e and $\kappa > 0$, take η from the hypothesis and $t = \eta/(2(|e|+1)) > 0$; then $|te| < \eta$, and since $\gamma = 0$ the hypothesis at $h = te$ gives $\frac{1}{2}t^2q(e) \leq \kappa t^2|e|^2$, i.e. $q(e) \leq 2\kappa|e|^2$. If $q(e) > 0$, choose $\kappa = q(e)/(4(|e|^2+1))$: then $q(e) \leq q(e)\frac{|e|^2}{2(|e|^2+1)} < q(e)$, a contradiction. So $q(e) \leq 0$ for every e .

Positive semidefiniteness. $-G$ is symmetric since G is, and for every $x \in \mathbb{R}^n$, $x^T(-G)x = -\langle Gx, x \rangle = -q(x) \geq 0$. Hence $-G$ is positive semidefinite. $\square$

2.18. quadratic_test_subsolution.

Lemma 2.18 (Quadratic test functions). *Let $n \geq 0$, $\Omega \subseteq \mathbb{R}^n$, $L : \mathbb{R}^n \times \mathbb{R}^{n \times n} \rightarrow \mathbb{R}$ and $f, u : \mathbb{R}^n \rightarrow \mathbb{R}$, and assume that u is a viscosity subsolution of $L(\nabla u, \nabla^2 u) = f$ in Ω in the sense of *IsViscositySubsolution*. Let $z \in \Omega$, $c \in \mathbb{R}$, $a \in \mathbb{R}^n$ and let A be a symmetric real $n \times n$ matrix. Put*

$$P(y) = c + \langle a, y - z \rangle + \frac{1}{2} \langle A(y - z), y - z \rangle \quad (y \in \mathbb{R}^n),$$

where $A(y - z)$ is the matrix-vector product. If $u - P$ attains a local maximum at z (i.e. $u(y) - P(y) \leq u(z) - P(z)$ for all y in a neighbourhood of z in $\mathbb{R}^n$), then

$$L(a, A) \leq f(z).$$

Proof. We check that P is an admissible test function and compute $\nabla P(z)$ and the Hessian $\nabla^2 P(z)$ of [EuclidHessian](#).

Symmetry. Since A is symmetric, $\langle Ah, k \rangle = \sum_{i,j} A_{ij} h_j k_i = \sum_{i,j} A_{ji} h_j k_i = \langle h, Ak \rangle$ for all $h, k \in \mathbb{R}^n$.

Regularity. In coordinates P is a polynomial of degree at most 2 in $y_1, \dots, y_n$, hence C^∞ on $\mathbb{R}^n$; in particular P is C^2 on Ω .

First derivative. For $y, h \in \mathbb{R}^n$, expanding bilinearly,

$P(y+h) - P(y) = \langle a, h \rangle + \frac{1}{2} \langle Ah, y-z \rangle + \frac{1}{2} \langle A(y-z), h \rangle + \frac{1}{2} \langle Ah, h \rangle = \langle a + A(y-z), h \rangle + \frac{1}{2} \langle Ah, h \rangle$, using $\langle Ah, y-z \rangle = \langle h, A(y-z) \rangle$. Since $|\frac{1}{2} \langle Ah, h \rangle| \leq \frac{1}{2} \|A\| |h|^2 = o(|h|)$, P is differentiable at y with $DP(y)h = \langle a + A(y-z), h \rangle$. Hence the gradient is $\nabla P(y) = a + A(y-z)$ and in particular $\nabla P(z) = a$.

Second derivative. By the previous paragraph, $DP(y) = \langle a, \cdot \rangle + \langle A(y-z), \cdot \rangle$ for every y , so $DP(y+k) - DP(y) = \langle Ak, \cdot \rangle$, which is linear in k . Hence $y \mapsto DP(y)$ is differentiable everywhere with $D^2 P(y)(k)(h) = \langle Ak, h \rangle$. By [EuclidHessian](#),

$$(\nabla^2 P(z))_{ij} = D^2 P(z)(e_i)(e_j) = \langle Ae_i, e_j \rangle = A_{ji} = A_{ij},$$

so $\nabla^2 P(z) = A$.

Conclusion. We have $z \in \Omega$, P is C^2 on Ω and $u - P$ has a local maximum at z . Clause (ii) of [IsViscositySubsolution](#) with $x_0 = z$ and $\psi = P$ gives $L(\nabla P(z), \nabla^2 P(z)) \leq f(z)$, that is, $L(a, A) \leq f(z)$. $\square$

2.19. supConvolution_jet_transfer.

Lemma 2.19 (Transfer of second-order upper expansions through the sup-convolution). *Let $n \geq 0$, $\Omega \subseteq \mathbb{R}^n$, and let $L : \mathbb{R}^n \times \mathbb{R}^{n \times n} \rightarrow \mathbb{R}$ be such that $(\xi, X) \mapsto L(\xi, X)$ is continuous on $\mathbb{R}^n \times \mathbb{S}^n$ ($\mathbb{S}^n$ the symmetric matrices, with the relative topology). Let $f, u : \mathbb{R}^n \rightarrow \mathbb{R}$ with u a viscosity subsolution of $L(\nabla u, \nabla^2 u) = f$ in Ω ([IsViscositySubsolution](#)). Let $K \subseteq \Omega$ be compact, $\varepsilon > 0$, and $u^\varepsilon = u_K^\varepsilon$ the sup-convolution of [SupConvolution](#). Let $x \in \mathbb{R}^n$ and let $\hat{x}$ be an interior point of K with*

$$u^\varepsilon(x) = u(\hat{x}) - \frac{|x - \hat{x}|^2}{2\varepsilon}.$$

Let $a \in \mathbb{R}^n$ and $A \in \mathbb{S}^n$, and assume that for every $\delta > 0$ the inequality

$$u^\varepsilon(z) \leq u^\varepsilon(x) + \langle a, z - x \rangle + \frac{1}{2} \langle A(z - x), z - x \rangle + \delta |z - x|^2$$

holds for all z in some neighbourhood of x . Then $L(a, A) \leq f(\hat{x})$.

Proof. Preliminaries. K is nonempty since $\hat{x} \in K$, and u is upper semicontinuous on Ω by clause (i) of [IsViscositySubsolution](#), hence on $K \subseteq \Omega$. So [supConvolution_properties](#)(i) applies: $u(y) - |z - y|^2/(2\varepsilon) \leq u^\varepsilon(z)$ for all $z \in \mathbb{R}^n$, $y \in K$.

Step 1. Fix $\delta > 0$. Choose $\rho_1 > 0$ such that the assumed inequality (with this δ) holds for $|z - x| < \rho_1$, and $\rho_2 > 0$ with $B(\hat{x}, \rho_2) \subseteq K$ ($\hat{x}$ is interior). Let $\rho = \min(\rho_1, \rho_2)$.

Step 2 (touching u from above at $\hat{x}$). Let $z' \in B(\hat{x}, \rho)$; then $z' \in K$. Put $z = z' + (x - \hat{x})$, so $z - x = z' - \hat{x}$, $|z - x| < \rho_1$ and $z - z' = x - \hat{x}$. By [supConvolution_properties](#)(i) applied at the point z with $y = z'$,

$$u(z') \leq u^\varepsilon(z) + \frac{|x - \hat{x}|^2}{2\varepsilon}.$$

By the assumed inequality at z and the hypothesis $u^\varepsilon(x) + \frac{|x - \hat{x}|^2}{2\varepsilon} = u(\hat{x})$,

$$u(z') \leq u(\hat{x}) + \langle a, z' - \hat{x} \rangle + \frac{1}{2} \langle A(z' - \hat{x}), z' - \hat{x} \rangle + \delta |z' - \hat{x}|^2 = P_\delta(z'),$$

where $P_\delta(y) = u(\hat{x}) + \langle a, y - \hat{x} \rangle + \frac{1}{2} \langle (A + 2\delta I)(y - \hat{x}), y - \hat{x} \rangle$ (note $\frac{1}{2} \langle 2\delta I h, h \rangle = \delta |h|^2$). Since $P_\delta(\hat{x}) = u(\hat{x})$, we get $u(z') - P_\delta(z') \leq 0 = u(\hat{x}) - P_\delta(\hat{x})$ for all $z' \in B(\hat{x}, \rho)$: $u - P_\delta$ attains a local maximum at $\hat{x}$.

Step 3. $\hat{x} \in K \subseteq \Omega$ and $A + 2\delta I$ is symmetric. By [quadratic_test_subsolution](#) (with $z = \hat{x}$, $c = u(\hat{x})$, a , and the symmetric matrix $A + 2\delta I$), $L(a, A + 2\delta I) \leq f(\hat{x})$.

Step 4 ($\delta \rightarrow 0$). The points $(a, A + 2\delta I)$, $\delta > 0$, lie in $\mathbb{R}^n \times \mathbb{S}^n$ and converge to $(a, A) \in \mathbb{R}^n \times \mathbb{S}^n$ as $\delta \rightarrow 0^+$. By continuity of L on $\mathbb{R}^n \times \mathbb{S}^n$, $L(a, A + 2\delta I) \rightarrow L(a, A)$. Passing to the limit in $L(a, A + 2\delta I) \leq f(\hat{x})$ gives $L(a, A) \leq f(\hat{x})$. $\square$

2.20. disjoint_sum_quadratic_strict.

Lemma 2.20 (Sum in disjoint variables of infinity-Laplace subsolutions against a strict quadratic maximum). *Let $n, m \geq 0$, write $\text{fst} : \mathbb{R}^{n+m} \rightarrow \mathbb{R}^n$ and $\text{snd} : \mathbb{R}^{n+m} \rightarrow \mathbb{R}^m$ for the block maps of [EuclidFst](#), [EuclidSnd](#), and $L_\infty(\xi, X) = -\langle X\xi, \xi \rangle$ for the operator of [InfLaplaceOperator](#) (in each dimension). Let $\mathcal{U} \subseteq \mathbb{R}^n$, $\mathcal{V} \subseteq \mathbb{R}^m$, let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be continuous on $\mathcal{U}$ and $g : \mathbb{R}^m \rightarrow \mathbb{R}$ continuous on $\mathcal{V}$ (relative to these sets), and let $u : \mathbb{R}^n \rightarrow \mathbb{R}$, $v : \mathbb{R}^m \rightarrow \mathbb{R}$ be viscosity subsolutions ([IsViscositySubsolution](#)) of $L_\infty(\nabla u, \nabla^2 u) = f$ in $\mathcal{U}$ and of $L_\infty(\nabla v, \nabla^2 v) = g$ in $\mathcal{V}$, respectively. Let $z_0 \in \mathbb{R}^{n+m}$ and $r > 0$ with*

$$\overline{B}(z_0, r) \subseteq \{z \in \mathbb{R}^{n+m} : \text{fst } z \in \mathcal{U}, \text{snd } z \in \mathcal{V}\},$$

let $q_0 \in \mathbb{R}$, $c \in \mathbb{R}^{n+m}$, let P be a symmetric real $(n+m) \times (n+m)$ matrix, and $Q(z) = q_0 + \langle c, z - z_0 \rangle + \frac{1}{2} \langle P(z - z_0), z - z_0 \rangle$. Put $x_0 = \text{fst } z_0$, $y_0 = \text{snd } z_0$ and assume the strict maximum

$$u(\text{fst } z) + v(\text{snd } z) - Q(z) < u(x_0) + v(y_0) - q_0 \quad \text{for all } z \in \overline{B}(z_0, r) \setminus \{z_0\}.$$

Then $L_\infty(c, P) \leq f(x_0) + g(y_0)$, i.e. $-\langle Pc, c \rangle \leq f(x_0) + g(y_0)$.

Proof. Relation to the paper's proof. The paper proves Theorem infsuper by applying the Theorem on Sums (Lemma TOS, [CIL, Theorem 3.2]) to $u(x) + v(y) - \varphi(x, y)$, which produces jets (ξ, X_ε) of u at x_0 and (η, Y_ε) of v at y_0 with $(\xi, \eta) = \nabla \varphi$ and the block inequality (2ndest) $\text{diag}(X_\varepsilon, Y_\varepsilon) \leq Z + \varepsilon Z^2$; multiplying (2ndest) by $\nabla \varphi$ on both sides gives $\langle X_\varepsilon \xi, \xi \rangle + \langle Y_\varepsilon \eta, \eta \rangle \leq \langle Z \nabla \varphi, \nabla \varphi \rangle + C\varepsilon$, and the two subsolution inequalities conclude. The sup-convolution/Jensen/Alexandrov argument below is exactly the standard proof of the Theorem on Sums (Crandall-Ishii-Lions, Theorem 3.2 and its appendix: sup-convolution, Jensen's lemma, Alexandrov's theorem), specialized to *disjoint* variables, just as in the common-variables node [sum_subsolution_quadratic_strict](#). It produces at one point z a genuine second-order jet (a, A) of u^ε at $\text{fst } z$ and (b, B) of v^ε at $\text{snd } z$ with $\xi := a \oplus b = c + P(z - z_0) - p$ and the block inequality $\text{diag}(A, B) \leq P$, the analogue of (2ndest); evaluating it at ξ is the paper's multiplication of (2ndest) by the gradient, and the jets are transferred to u, v at nearby points by [supConvolution_jet_transfer](#) (no closure of semijets is needed).

Notation and block calculus. For $x \in \mathbb{R}^n$, $y \in \mathbb{R}^m$ let $x \oplus y \in \mathbb{R}^{n+m}$ be the concatenation, so $\text{fst}(x \oplus y) = x$, $\text{snd}(x \oplus y) = y$ and $z = \text{fst } z \oplus \text{snd } z$. The maps fst, snd are linear (they select coordinates), and by [euclidSplit_norm_sq](#), $|z|^2 = |\text{fst } z|^2 + |\text{snd } z|^2$; by polarization $(\langle z, w \rangle = \frac{1}{2}(|z + w|^2 - |z|^2 - |w|^2))$ and linearity) also

$$(0) \quad \langle z, w \rangle = \langle \text{fst } z, \text{fst } w \rangle + \langle \text{snd } z, \text{snd } w \rangle, \quad |\text{fst } z| \leq |z|, \quad |\text{snd } z| \leq |z|.$$

Write $h' = \text{fst } h$, $h'' = \text{snd } h$. Let $K_1 = \overline{B}(x_0, r) \subseteq \mathbb{R}^n$ and $K_2 = \overline{B}(y_0, r) \subseteq \mathbb{R}^m$ (compact, nonempty), and $K = \overline{B}(z_0, r)$. Then $K_1 \subseteq \mathcal{U}$: for $x \in K_1$ the point $x \oplus y_0$ satisfies $|x \oplus y_0 - z_0|^2 = |x - x_0|^2 + 0 \leq r^2$, so it lies in K , and hence $x = \text{fst}(x \oplus y_0) \in \mathcal{U}$. Likewise $K_2 \subseteq \mathcal{V}$ (use $x_0 \oplus y$). Also $\text{fst}(K) \subseteq K_1$ and $\text{snd}(K) \subseteq K_2$ by (0). By clause (i) of [IsViscositySubsolution](#), u is upper semicontinuous on $\mathcal{U} \supseteq K_1$ and v on $\mathcal{V} \supseteq K_2$; so u attains a maximum M_u on K_1 and v a maximum M_v on K_2 (upper semicontinuous functions on nonempty compact sets attain their maxima). Let $M_Q \geq \sup_K |Q|$ and $\|P\|$ the operator norm, so $\langle Ph, h \rangle \leq \|P\| |h|^2$ and $|Ph| \leq \|P\| |h|$. Since P is symmetric, for all z, h ,

$$(1) \quad Q(z + h) = Q(z) + \langle c + P(z - z_0), h \rangle + \frac{1}{2} \langle Ph, h \rangle.$$

Let $m_0 = u(x_0) + v(y_0) - q_0 = u(x_0) + v(y_0) - Q(z_0)$. For $\varepsilon > 0$ let $u^\varepsilon = u_{K_1}^\varepsilon$, $v^\varepsilon = v_{K_2}^\varepsilon$ ([SupConvolution](#)) and

$$W_\varepsilon(z) = u^\varepsilon(\text{fst } z) + v^\varepsilon(\text{snd } z) - Q(z).$$

By [supConvolution_properties](#) (for u on K_1 and v on K_2): (i) $u(y) - \frac{|x-y|^2}{2\varepsilon} \leq u^\varepsilon(x)$ for $y \in K_1$; (ii) for each x there is $\hat{x} \in K_1$ with $u^\varepsilon(x) = u(\hat{x}) - \frac{|x-\hat{x}|^2}{2\varepsilon}$; (iii) $F_u^\varepsilon(x) = u^\varepsilon(x) + \frac{|x|^2}{2\varepsilon}$ is

convex on $\mathbb{R}^n$; and likewise for v on K_2 with F_v^ε convex on $\mathbb{R}^m$. By (i) with $x = y = x_0$ and $x = y = y_0$,

$$(2) \quad W_\varepsilon(z_0) \geq m_0.$$

Concentration. Q is continuous and the strict maximum hypothesis is exactly that of `disjoint_supConvolut` (with u upper semicontinuous on K_1 , v on K_2). Hence for every $\rho > 0$ there are $\beta_\rho, \varepsilon_\rho > 0$ with $W_\varepsilon(z) \leq m_0 - \beta_\rho$ for all $\varepsilon \in (0, \varepsilon_\rho)$ and $z \in K$ with $|z - z_0| \geq \rho$.

Choice of parameters. Fix $\sigma > 0$. The map $\xi \mapsto L_\infty(\xi, P) = -\langle P\xi, \xi \rangle$ is continuous (a polynomial), f is continuous on $\mathcal{U}$ at $x_0 \in K_1 \subseteq \mathcal{U}$ and g on $\mathcal{V}$ at $y_0 \in K_2 \subseteq \mathcal{V}$. Choose $\rho \in (0, r/2]$ such that

$$(3)$$

$$L_\infty(\xi, P) \geq L_\infty(c, P) - \sigma \text{ if } |\xi - c| \leq (\|P\| + 1)\rho; \quad f(x') \leq f(x_0) + \sigma \text{ if } x' \in \mathcal{U}, |x' - x_0| < 2\rho; \quad g(y') \leq g(y_0) + \sigma$$

Let $\beta = \beta_\rho$, ε_ρ as in the concentration step, $C_0 = M_u + M_v + M_Q - m_0 + \beta$. Choose $\varepsilon \in (0, \varepsilon_\rho)$ with $2\varepsilon \max(C_0, 0) < \rho^2$ and δ with $0 < \delta < \min(\rho, \frac{\beta}{2r})$.

Semiconvexity of W_ε . Let $\lambda = \|P\|$ and $C_W = \frac{1}{\varepsilon} + \lambda \geq 0$. Using $|z|^2 = |\text{fst } z|^2 + |\text{snd } z|^2$,

$$W_\varepsilon(z) + \frac{C_W}{2}|z|^2 = F_u^\varepsilon(\text{fst } z) + F_v^\varepsilon(\text{snd } z) + \left(\frac{\lambda}{2}|z|^2 - Q(z)\right).$$

The first two terms are convex on $\mathbb{R}^{n+m}$, being convex functions composed with linear maps ($F(\ell(tz_1 + (1-t)z_2)) = F(t\ell z_1 + (1-t)\ell z_2) \leq tF(\ell z_1) + (1-t)F(\ell z_2)$). The third is convex by `quadratic_convex_part` (in dimension $n+m$, with $\langle Ph, h \rangle \leq \lambda|h|^2$). So $W_\varepsilon + \frac{C_W}{2}|\cdot|^2$ is convex on $\mathbb{R}^{n+m}$.

Jensen. If $|z - z_0| = r$ then $z \in K$ and $|z - z_0| \geq \rho$, so by concentration and (2), $W_\varepsilon(z) + 2\delta r \leq m_0 - \beta + 2\delta r < m_0 \leq W_\varepsilon(z_0)$. By `jensen_positive_measure` (in dimension $n+m$, with $W = W_\varepsilon$, $C = C_W$, $x_0 = x_* = z_0$), the set K_δ of $z \in B(z_0, r)$ for which some $p \in \mathbb{R}^{n+m}$ with $|p| \leq \delta$ satisfies $W_\varepsilon(z') + \langle p, z' \rangle \leq W_\varepsilon(z) + \langle p, z \rangle$ for all $z' \in K$ has $\text{vol}(K_\delta) > 0$.

Alexandrov. By `alexandrov_convex` applied to the convex F_u^ε on $\mathbb{R}^n$ and F_v^ε on $\mathbb{R}^m$, there are null sets $N_u \subseteq \mathbb{R}^n$, $N_v \subseteq \mathbb{R}^m$ outside of which these functions have second-order expansions. By `euclidSplit_preimage_null`, $N = \{z : \text{fst } z \in N_u\} \cup \{z : \text{snd } z \in N_v\}$ is null in $\mathbb{R}^{n+m}$. As $\text{vol}(K_\delta) \leq \text{vol}(K_\delta \setminus N) + \text{vol}(N)$, $K_\delta \setminus N \neq \emptyset$; pick z in it with witness p , $|p| \leq \delta$, and put $\bar{x} = \text{fst } z$, $\bar{y} = \text{snd } z$. As $\bar{x} \notin N_u$, there are $a_0 \in \mathbb{R}^n$ and a symmetric A_0 with $F_u^\varepsilon(x) - F_u^\varepsilon(\bar{x}) - \langle a_0, x - \bar{x} \rangle - \frac{1}{2}\langle A_0(x - \bar{x}), x - \bar{x} \rangle = o(|x - \bar{x}|^2)$. By `expansion_sub_half_sq_norm` (with $w = u^\varepsilon$), $a = a_0 - \frac{1}{\varepsilon}\bar{x}$ and the symmetric matrix $A = A_0 - \frac{1}{\varepsilon}I$ satisfy: for every $\kappa > 0$, for all x near $\bar{x}$,

$$(4) \quad |u^\varepsilon(x) - u^\varepsilon(\bar{x}) - \langle a, x - \bar{x} \rangle - \frac{1}{2}\langle A(x - \bar{x}), x - \bar{x} \rangle| \leq \kappa|x - \bar{x}|^2.$$

Likewise there are $b \in \mathbb{R}^m$ and a symmetric B with the same property for v^ε at $\bar{y}$. (4')

Location of z , $\hat{x}$, $\hat{y}$. By the maximality defining K_δ (with $z' = z_0$) and (2), $W_\varepsilon(z) \geq W_\varepsilon(z_0) - \langle p, z - z_0 \rangle \geq m_0 - \delta r > m_0 - \beta$ (as $|z - z_0| < r$ and $2\delta r < \beta$). By concentration, $|z - z_0| < \rho$; hence $|\bar{x} - x_0|, |\bar{y} - y_0| < \rho$ by (0). Take $\hat{x} \in K_1$, $\hat{y} \in K_2$ as in (ii) for $u^\varepsilon(\bar{x})$, $v^\varepsilon(\bar{y})$. Then

$$\frac{|\bar{x} - \hat{x}|^2 + |\bar{y} - \hat{y}|^2}{2\varepsilon} = u(\hat{x}) + v(\hat{y}) - Q(z) - W_\varepsilon(z) \leq M_u + M_v + M_Q - m_0 + \beta = C_0,$$

so $|\bar{x} - \hat{x}|, |\bar{y} - \hat{y}| \leq \sqrt{2\varepsilon \max(C_0, 0)} < \rho$ and $|\hat{x} - x_0|, |\hat{y} - y_0| < 2\rho \leq r$. Thus $\hat{x} \in B(x_0, r)$ and $\hat{y} \in B(y_0, r)$ are interior points of K_1 and K_2 .

First- and second-order conditions (block form of (2ndest)). Let D be the block-diagonal $(n+m) \times (n+m)$ matrix with $D(h) = Ah' \oplus Bh''$ (entries of A in the first n rows and columns, of B in the last m , zero elsewhere); D is symmetric and by (0) $\langle Dh, h \rangle = \langle Ah', h' \rangle + \langle Bh'', h'' \rangle$. Let $\xi = a \oplus b$, so $\langle \xi, h \rangle = \langle a, h' \rangle + \langle b, h'' \rangle$ by (0). Put $\gamma = \xi - c - P(z - z_0) + p$ and $G = D - P$ (symmetric). Fix $\kappa > 0$. Since $z \in B(z_0, r)$, for all sufficiently small $|h|$ we have $z + h \in K$, hence $W_\varepsilon(z + h) + \langle p, h \rangle \leq W_\varepsilon(z)$, and (4), (4') apply at $\bar{x} + h'$, $\bar{y} + h''$ (as $|h'|, |h''| \leq |h|$). Using (1), (4), (4') and $|h'|^2 + |h''|^2 = |h|^2$,

$$0 \geq W_\varepsilon(z + h) - W_\varepsilon(z) + \langle p, h \rangle \geq \langle \gamma, h \rangle + \frac{1}{2}\langle Gh, h \rangle - \kappa|h|^2.$$

Thus for every $\kappa > 0$ there is $\eta > 0$ with $\langle \gamma, h \rangle + \frac{1}{2} \langle Gh, h \rangle \leq \kappa |h|^2$ for $|h| < \eta$. By [quadratic_upper_bound_cond](#), $\gamma = 0$ and $\langle Ge, e \rangle \leq 0$ for all e , i.e.

$$\xi = a \oplus b = c + P(z - z_0) - p, \quad \langle Ae', e' \rangle + \langle Be'', e'' \rangle = \langle De, e \rangle \leq \langle Pe, e \rangle \text{ for all } e \in \mathbb{R}^{n+m}.$$

Taking $e = \xi$ (so $e' = a$, $e'' = b$) — the paper's multiplication of (2ndest) by the gradient on both sides —

$$(5) \quad \langle Aa, a \rangle + \langle Bb, b \rangle \leq \langle P\xi, \xi \rangle.$$

Subsolution inequalities via jet transfer. L_∞ is continuous on $\mathbb{R}^n \times \mathbb{R}^{n \times n}$ (a polynomial), in particular on $\mathbb{R}^n \times \mathbb{S}^n$. By (4), for every $\kappa > 0$, $u^\varepsilon(x) \leq u^\varepsilon(\bar{x}) + \langle a, x - \bar{x} \rangle + \frac{1}{2} \langle A(x - \bar{x}), x - \bar{x} \rangle + \kappa |x - \bar{x}|^2$ near $\bar{x}$; also $u^\varepsilon(\bar{x}) = u(\hat{x}) - \frac{|\bar{x} - \hat{x}|^2}{2\varepsilon}$ with $\hat{x}$ interior to the compact set $K_1 \subseteq \mathcal{U}$, and A is symmetric. By [supConvolution_jet_transfer](#) (with $\Omega = \mathcal{U}$, $L = L_\infty$, $K = K_1$, $x = \bar{x}$), $L_\infty(a, A) = -\langle Aa, a \rangle \leq f(\hat{x})$. In the same way (with $\mathcal{V}$, K_2 , $\bar{y}$, $\hat{y}$, v , g), $-\langle Bb, b \rangle \leq g(\hat{y})$.

The chain and the limit. By (5) and the two subsolution inequalities,

$$L_\infty(\xi, P) = -\langle P\xi, \xi \rangle \leq -\langle Aa, a \rangle - \langle Bb, b \rangle \leq f(\hat{x}) + g(\hat{y}).$$

Now $|\xi - c| = |P(z - z_0) - p| \leq \|P\|\rho + \delta \leq (\|P\| + 1)\rho$, $\hat{x} \in K_1 \subseteq \mathcal{U}$ with $|\hat{x} - x_0| < 2\rho$, and $\hat{y} \in K_2 \subseteq \mathcal{V}$ with $|\hat{y} - y_0| < 2\rho$. By (3),

$$L_\infty(c, P) - \sigma \leq L_\infty(\xi, P) \leq f(\hat{x}) + g(\hat{y}) \leq f(x_0) + g(y_0) + 2\sigma.$$

As $\sigma > 0$ was arbitrary, $L_\infty(c, P) \leq f(x_0) + g(y_0)$. $\square$

2.21. infLaplace_supersolution_iff_neg.

Lemma 2.21 (Supersolutions of the infinity-Laplace equation via negation). *Let $n \geq 0$, $\Omega \subseteq \mathbb{R}^n$ and $f, u : \mathbb{R}^n \rightarrow \mathbb{R}$, and let $L_\infty(\xi, X) = -\langle X\xi, \xi \rangle$ be the operator of [InfLaplaceOperator](#). Then u is a viscosity supersolution of $L_\infty(\nabla u, \nabla^2 u) = f$ in Ω ([IsViscositySupersolution](#)) if and only if $-u$ is a viscosity subsolution of $L_\infty(\nabla(-u), \nabla^2(-u)) = -f$ in Ω ([IsViscositySubsolution](#)).*

Proof. Preliminary identities. Let $\psi : \mathbb{R}^n \rightarrow \mathbb{R}$ and $x \in \mathbb{R}^n$.

- (a) $D(-\psi)(y) = -D\psi(y)$ for every y : if ψ is differentiable at y this is linearity of the derivative, and otherwise $-\psi$ is not differentiable at y either (as $\psi = -(-\psi)$), so both sides are 0 by the convention of [EuclidHessian](#). Hence $\nabla(-\psi)(x) = -\nabla\psi(x)$.
- (b) Applying (a) to the map $y \mapsto D\psi(y)$ (same argument for vector-valued maps), $D^2(-\psi)(x) = D(-D\psi)(x) = -D^2\psi(x)$, so by [EuclidHessian](#) $\nabla^2(-\psi)(x) = -\nabla^2\psi(x)$.
- (c) For all ξ, X : $L_\infty(-\xi, -X) = -\langle (-X)(-\xi), -\xi \rangle = -(-\langle X\xi, \xi \rangle) = -L_\infty(\xi, X)$.
- (d) ψ is C^2 on Ω iff $-\psi$ is (the C^2 functions on Ω form a vector space).
- (e) $u - \psi$ has a local minimum at x_0 iff $(-u) - (-\psi) = -(u - \psi)$ has a local maximum at x_0 .
- (f) u is lower semicontinuous on Ω iff $-u$ is upper semicontinuous on Ω : for $x \in \Omega$, " $t < u(x) \Rightarrow u > t$ near x in Ω " is, with $s = -t$, " $s > -u(x) \Rightarrow -u < s$ near x in Ω ".

Combining (a)–(c), for every ψ and x_0 ,

$$(*) \quad L_\infty(\nabla(-\psi)(x_0), \nabla^2(-\psi)(x_0)) = L_\infty(-\nabla\psi(x_0), -\nabla^2\psi(x_0)) = -L_\infty(\nabla\psi(x_0), \nabla^2\psi(x_0)).$$

$(\Rightarrow)$. Assume u is a supersolution. Clause (i) for $-u$ holds by (f). For clause (ii), let $x_0 \in \Omega$ and φ be C^2 on Ω with $(-u) - \varphi$ having a local maximum at x_0 . Put $\psi = -\varphi$; by (d) ψ is C^2 on Ω , and $u - \psi = -((-u) - \varphi)$ has a local minimum at x_0 by (e). The supersolution property gives $f(x_0) \leq L_\infty(\nabla\psi(x_0), \nabla^2\psi(x_0))$. By (*) with $-\psi = \varphi$, $L_\infty(\nabla\varphi(x_0), \nabla^2\varphi(x_0)) = -L_\infty(\nabla\psi(x_0), \nabla^2\psi(x_0)) \leq -f(x_0)$, which is clause (ii) for $-u$ with right-hand side $-f$.

$(\Leftarrow)$. Assume $-u$ is a subsolution with right-hand side $-f$. Clause (i) for u holds by (f). Let $x_0 \in \Omega$ and ψ be C^2 on Ω with $u - \psi$ having a local minimum at x_0 . Then $-\psi$ is C^2 on Ω (d) and $(-u) - (-\psi)$ has a local maximum at x_0 (e), so $L_\infty(\nabla(-\psi)(x_0), \nabla^2(-\psi)(x_0)) \leq -f(x_0)$. By (*) this reads $-L_\infty(\nabla\psi(x_0), \nabla^2\psi(x_0)) \leq -f(x_0)$, i.e. $f(x_0) \leq L_\infty(\nabla\psi(x_0), \nabla^2\psi(x_0))$, which is clause (ii) of the supersolution property. $\square$

2.22. slice_left_admissible.

Lemma 2.22 (Slices of compactly contained sets, first coordinate, vertex removed). *Let $(\mathbf{X}, d_{\mathbf{X}})$, $(\mathbf{Y}, d_{\mathbf{Y}})$ be metric spaces and equip $\mathbf{X} \times \mathbf{Y}$ with the ℓ^1 metric $d_1((x_1, y_1), (x_2, y_2)) = d_{\mathbf{X}}(x_1, x_2) + d_{\mathbf{Y}}(y_1, y_2)$. Let $\mathcal{U} \subseteq \mathbf{X}$, $\mathcal{V} \subseteq \mathbf{Y}$, and let $\mathcal{O} \subseteq \mathbf{X} \times \mathbf{Y}$ be open and bounded (for d_1) with $\overline{\mathcal{O}}$ compact and $\overline{\mathcal{O}} \subseteq \mathcal{U} \times \mathcal{V}$. Fix $y_0 \in \mathbf{Y}$ and $\hat{x} \in \mathbf{X}$ and set*

$$\mathcal{O}_{\mathbf{X}} = \{x \in \mathbf{X} : (x, y_0) \in \mathcal{O}\} \setminus \{\hat{x}\}.$$

Then $\mathcal{O}_{\mathbf{X}}$ is open and bounded in $\mathbf{X}$, $\overline{\mathcal{O}_{\mathbf{X}}}$ is compact, $\overline{\mathcal{O}_{\mathbf{X}}} \subseteq \mathcal{U}$, $\overline{\mathcal{O}_{\mathbf{X}}} \subseteq \{x : (x, y_0) \in \overline{\mathcal{O}}\}$, and

$$\partial \mathcal{O}_{\mathbf{X}} \subseteq \{x : (x, y_0) \in \partial \mathcal{O}\} \cup \{\hat{x}\}.$$

Proof. Let $\iota : \mathbf{X} \rightarrow \mathbf{X} \times \mathbf{Y}$, $\iota(x) = (x, y_0)$, and $A = \iota^{-1}(\mathcal{O}) = \{x : (x, y_0) \in \mathcal{O}\}$, so $\mathcal{O}_{\mathbf{X}} = A \setminus \{\hat{x}\}$. By the definition of d_1 , $d_1(\iota(x), \iota(x')) = d_{\mathbf{X}}(x, x') + d_{\mathbf{Y}}(y_0, y_0) = d_{\mathbf{X}}(x, x')$, so ι is an isometry, in particular continuous, and $\pi_1 \circ \iota = \text{id}_{\mathbf{X}}$ for the continuous projection $\pi_1(x, y) = x$.

Open. A is open as the preimage of the open set $\mathcal{O}$ under the continuous map ι ; $\{\hat{x}\}$ is closed in the metric space $\mathbf{X}$; so $\mathcal{O}_{\mathbf{X}} = A \setminus \{\hat{x}\}$ is open.

Bounded. Choose C with $d_1(z, z') \leq C$ for $z, z' \in \mathcal{O}$. For $x, x' \in \mathcal{O}_{\mathbf{X}} \subseteq A$, $d_{\mathbf{X}}(x, x') = d_1(\iota(x), \iota(x')) \leq C$.

Closure inside the slice of $\overline{\mathcal{O}}$. $\overline{\mathcal{O}_{\mathbf{X}}} \subseteq \overline{A} \subseteq \iota^{-1}(\overline{\mathcal{O}})$, the last inclusion because $\iota^{-1}(\overline{\mathcal{O}})$ is closed and contains A .

Closure inside $\mathcal{U}$. If $x \in \overline{\mathcal{O}_{\mathbf{X}}}$ then $(x, y_0) \in \overline{\mathcal{O}} \subseteq \mathcal{U} \times \mathcal{V}$, so $x \in \mathcal{U}$.

Compact closure. As $\pi_1 \circ \iota = \text{id}$ with π_1, ι continuous and $\mathbf{X} \times \mathbf{Y}$ Hausdorff, ι is a closed embedding (its image is the closed set $\{z : \iota(\pi_1 z) = z\}$ and π_1 restricted to it is a continuous inverse). Hence $\iota^{-1}(\overline{\mathcal{O}})$ is compact, and its closed subset $\overline{\mathcal{O}_{\mathbf{X}}}$ is compact.

Boundary. Let $x \in \partial \mathcal{O}_{\mathbf{X}}$. Since $\mathcal{O}_{\mathbf{X}}$ is open, $\partial \mathcal{O}_{\mathbf{X}} = \overline{\mathcal{O}_{\mathbf{X}}} \setminus \mathcal{O}_{\mathbf{X}}$, so $x \in \overline{\mathcal{O}_{\mathbf{X}}}$ and $x \notin \mathcal{O}_{\mathbf{X}}$. If $x = \hat{x}$ we are done. Otherwise $x \neq \hat{x}$ and $x \notin A \setminus \{\hat{x}\}$ give $x \notin A$; also $x \in \overline{\mathcal{O}_{\mathbf{X}}} \subseteq \overline{A}$. As A is open, $x \in \overline{A} \setminus A = \partial A$. Finally $\partial A = \partial(\iota^{-1}(\mathcal{O})) \subseteq \iota^{-1}(\partial \mathcal{O})$: indeed $x \in \overline{A}$ gives $\iota(x) \in \overline{\mathcal{O}}$ and $x \notin A$ gives $\iota(x) \notin \mathcal{O} = \text{int } \mathcal{O}$. Thus $(x, y_0) \in \partial \mathcal{O}$. $\square$

2.23. slice_right_admissible.

Lemma 2.23 (Slices of compactly contained sets, second coordinate). *Let $(\mathbf{X}, d_{\mathbf{X}})$, $(\mathbf{Y}, d_{\mathbf{Y}})$ be metric spaces and equip $\mathbf{X} \times \mathbf{Y}$ with the ℓ^1 metric $d_1((x_1, y_1), (x_2, y_2)) = d_{\mathbf{X}}(x_1, x_2) + d_{\mathbf{Y}}(y_1, y_2)$. Let $\mathcal{U} \subseteq \mathbf{X}$, $\mathcal{V} \subseteq \mathbf{Y}$, and let $\mathcal{O} \subseteq \mathbf{X} \times \mathbf{Y}$ be open and bounded (for d_1) with $\overline{\mathcal{O}}$ compact and $\overline{\mathcal{O}} \subseteq \mathcal{U} \times \mathcal{V}$. Fix $\hat{x} \in \mathbf{X}$ and set*

$$\mathcal{O}_{\mathbf{Y}} = \{y \in \mathbf{Y} : (\hat{x}, y) \in \mathcal{O}\}.$$

Then $\mathcal{O}_{\mathbf{Y}}$ is open and bounded in $\mathbf{Y}$, $\overline{\mathcal{O}_{\mathbf{Y}}}$ is compact, $\overline{\mathcal{O}_{\mathbf{Y}}} \subseteq \mathcal{V}$, $\overline{\mathcal{O}_{\mathbf{Y}}} \subseteq \{y : (\hat{x}, y) \in \overline{\mathcal{O}}\}$, and $\partial \mathcal{O}_{\mathbf{Y}} \subseteq \{y : (\hat{x}, y) \in \partial \mathcal{O}\}$.

Proof. Let $\iota : \mathbf{Y} \rightarrow \mathbf{X} \times \mathbf{Y}$, $\iota(y) = (\hat{x}, y)$, so that $\mathcal{O}_{\mathbf{Y}} = \iota^{-1}(\mathcal{O})$. By the definition of d_1 ,

$$d_1(\iota(y), \iota(y')) = d_{\mathbf{X}}(\hat{x}, \hat{x}) + d_{\mathbf{Y}}(y, y') = d_{\mathbf{Y}}(y, y'),$$

so ι is an isometry, in particular continuous. Moreover $\pi_2 \circ \iota = \text{id}_{\mathbf{Y}}$ for the continuous projection $\pi_2(x, y) = y$.

Open. $\mathcal{O}_{\mathbf{Y}} = \iota^{-1}(\mathcal{O})$ is the preimage of an open set under a continuous map.

Bounded. Since $\mathcal{O}$ is bounded there is C with $d_1(z, z') \leq C$ for all $z, z' \in \mathcal{O}$. For $y, y' \in \mathcal{O}_{\mathbf{Y}}$ we have $\iota(y), \iota(y') \in \mathcal{O}$, hence $d_{\mathbf{Y}}(y, y') = d_1(\iota(y), \iota(y')) \leq C$.

Closure inside the slice of $\overline{\mathcal{O}}$. $\iota^{-1}(\overline{\mathcal{O}})$ is closed (continuity of ι) and contains $\iota^{-1}(\mathcal{O})$, hence $\overline{\mathcal{O}_{\mathbf{Y}}} \subseteq \iota^{-1}(\overline{\mathcal{O}}) = \{y : (\hat{x}, y) \in \overline{\mathcal{O}}\}$.

Closure inside $\mathcal{V}$. If $y \in \overline{\mathcal{O}_{\mathbf{Y}}}$ then $(\hat{x}, y) \in \overline{\mathcal{O}} \subseteq \mathcal{U} \times \mathcal{V}$ by the previous step, so $y \in \mathcal{V}$.

Compact closure. Since $\pi_2 \circ \iota = \text{id}$ with π_2, ι continuous and $\mathbf{X} \times \mathbf{Y}$ Hausdorff, ι is a closed embedding: its image $\{(\hat{x}, y)\} = \{z : \iota(\pi_2 z) = z\}$ is closed (as the equalizer of two continuous maps into a Hausdorff space) and ι is a homeomorphism onto it with inverse π_2 . The preimage of a compact set under a closed embedding is compact, so $\iota^{-1}(\overline{\mathcal{O}})$ is compact. The set $\overline{\mathcal{O}_{\mathbf{Y}}}$ is a closed subset of it, hence compact.

Boundary. Let $y \in \partial\mathcal{O}_Y$. Then $y \in \overline{\mathcal{O}_Y}$, so $\iota(y) \in \overline{\mathcal{O}}$ as shown above. Also $y \notin \text{int } \mathcal{O}_Y = \mathcal{O}_Y$ (it is open), so $\iota(y) \notin \mathcal{O} = \text{int } \mathcal{O}$. Hence $\iota(y) = (\hat{x}, y) \in \overline{\mathcal{O}} \setminus \text{int } \mathcal{O} = \partial\mathcal{O}$. $\square$

2.24. supConvolution_concentration.

Lemma 2.24 (Concentration of the sup-convolutions near a strict maximum). *Let $n \geq 0$, $x_0 \in \mathbb{R}^n$, $r > 0$, $K = \overline{B}(x_0, r)$, let $u, v : \mathbb{R}^n \rightarrow \mathbb{R}$ be upper semicontinuous on K (relative to K), and let $Q : \mathbb{R}^n \rightarrow \mathbb{R}$ be continuous. Assume that $u + v - Q$ has a strict maximum over K at x_0 :*

$$u(y) + v(y) - Q(y) < m_0 := u(x_0) + v(x_0) - Q(x_0) \quad \text{for all } y \in K \setminus \{x_0\}.$$

For $\varepsilon > 0$ let $u^\varepsilon = u_K^\varepsilon$ and $v^\varepsilon = v_K^\varepsilon$ be the sup-convolutions of [SupConvolution](#). Then for every $\rho > 0$ there are $\beta > 0$ and $\varepsilon_0 > 0$ such that

$$u^\varepsilon(y) + v^\varepsilon(y) - Q(y) \leq m_0 - \beta \quad \text{for all } \varepsilon \in (0, \varepsilon_0) \text{ and all } y \in K \text{ with } |y - x_0| \geq \rho.$$

Proof. K is compact and nonempty. Since u, v are upper semicontinuous on K they attain their maxima on K ; let $M_u = \max_K u$, $M_v = \max_K v$, and let $M_Q \geq \sup_K |Q|$ (Q is continuous on the compact K). By [supConvolution_properties\(ii\)](#), for every $\varepsilon > 0$ and y there are $\hat{y}, \hat{y}' \in K$ with $u^\varepsilon(y) = u(\hat{y}) - \frac{|y - \hat{y}|^2}{2\varepsilon}$ and $v^\varepsilon(y) = v(\hat{y}') - \frac{|y - \hat{y}'|^2}{2\varepsilon}$.

Fix $\rho > 0$ and let $A_\rho = \{y \in K : |y - x_0| \geq \rho\}$, a compact subset of K . If $A_\rho = \emptyset$ the claim holds vacuously with $\beta = \varepsilon_0 = 1$. Otherwise $w := u + v - Q$ is upper semicontinuous on A_ρ and attains its maximum m_ρ there at some y_ρ ; as $|y_\rho - x_0| \geq \rho > 0$, $y_\rho \neq x_0$ and so $m_\rho < m_0$ by hypothesis. Put $\beta = \frac{1}{2}(m_0 - m_\rho) > 0$, so $m_\rho + \beta = m_0 - \beta$.

Local control. For each $y^* \in A_\rho$, upper semicontinuity of u and v on K at y^* and continuity of Q at y^* give $\eta(y^*) > 0$ such that $u(z) < u(y^*) + \beta/3$ and $v(z) < v(y^*) + \beta/3$ for $z \in K$ with $|z - y^*| < \eta(y^*)$, and $Q(z) > Q(y^*) - \beta/3$ for all z with $|z - y^*| < \eta(y^*)$. The open balls $B(y^*, \eta(y^*))$, $y^* \in A_\rho$, cover the compact set A_ρ , so by the Lebesgue number lemma there is $\lambda > 0$ such that for every $y \in A_\rho$ there is $y^* \in A_\rho$ with $B(y, \lambda) \subseteq B(y^*, \eta(y^*))$.

Choice of ε_0 . Let $C_0 = M_u + M_v + M_Q - m_0 + \beta$ and $\varepsilon_0 = \lambda^2 / (2(|C_0| + 1)) > 0$. Let $0 < \varepsilon < \varepsilon_0$ and $y \in A_\rho$, and suppose for contradiction that $u^\varepsilon(y) + v^\varepsilon(y) - Q(y) > m_0 - \beta$. With $\hat{y}, \hat{y}'$ as above,

$$\frac{|y - \hat{y}|^2}{2\varepsilon} + \frac{|y - \hat{y}'|^2}{2\varepsilon} = u(\hat{y}) + v(\hat{y}') - Q(y) - (u^\varepsilon(y) + v^\varepsilon(y) - Q(y)) < M_u + M_v + M_Q - m_0 + \beta = C_0 \leq |C_0|,$$

using $u(\hat{y}) \leq M_u$, $v(\hat{y}') \leq M_v$, $-Q(y) \leq M_Q$. Both squares are nonnegative, so $|y - \hat{y}|^2 < 2\varepsilon|C_0| < \lambda^2$ and likewise $|y - \hat{y}'|^2 < \lambda^2$; thus $\hat{y}, \hat{y}' \in B(y, \lambda)$. Choose y^* with $B(y, \lambda) \subseteq B(y^*, \eta(y^*))$; then $y, \hat{y}, \hat{y}' \in B(y^*, \eta(y^*))$ and $\hat{y}, \hat{y}' \in K$, so

$$u(\hat{y}) + v(\hat{y}') - Q(y) < u(y^*) + v(y^*) - Q(y^*) + \beta \leq m_\rho + \beta = m_0 - \beta.$$

Since $u^\varepsilon(y) + v^\varepsilon(y) - Q(y) \leq u(\hat{y}) + v(\hat{y}') - Q(y)$ (the subtracted squares are nonnegative), we get $u^\varepsilon(y) + v^\varepsilon(y) - Q(y) < m_0 - \beta$, a contradiction. Hence $u^\varepsilon(y) + v^\varepsilon(y) - Q(y) \leq m_0 - \beta$ for all $\varepsilon \in (0, \varepsilon_0)$ and $y \in A_\rho$. $\square$

2.25. sum_subsolution_quadratic_strict.

Lemma 2.25 (Sum of subsolutions against a strict quadratic maximum). *Let $n \geq 0$, $\Omega \subseteq \mathbb{R}^n$, $f, g : \mathbb{R}^n \rightarrow \mathbb{R}$ continuous on Ω (relative to Ω), and let $L : \mathbb{R}^n \times \mathbb{R}^{n \times n} \rightarrow \mathbb{R}$ be continuous on $\mathbb{R}^n \times \mathbb{S}^n$ ($\mathbb{S}^n$ the symmetric matrices), elliptic in the sense of [IsDegenerateElliptic](#) and subadditive in the sense of [IsSubadditiveOperator](#). Let $u, v : \mathbb{R}^n \rightarrow \mathbb{R}$ be viscosity subsolutions ([IsViscositySubsolution](#)) in Ω of $L(\nabla u, \nabla^2 u) = f$ and of $L(\nabla v, \nabla^2 v) = g$, respectively. Let $x_0 \in \mathbb{R}^n$ and $r > 0$ with $\overline{B}(x_0, r) \subseteq \Omega$, let $q_0 \in \mathbb{R}$, $c \in \mathbb{R}^n$, $P \in \mathbb{S}^n$, and*

$$Q(y) = q_0 + \langle c, y - x_0 \rangle + \frac{1}{2} \langle P(y - x_0), y - x_0 \rangle.$$

Assume that $u + v - Q$ has a strict maximum over $\overline{B}(x_0, r)$ at x_0 :

$$u(y) + v(y) - Q(y) < u(x_0) + v(x_0) - q_0 \quad \text{for all } y \in \overline{B}(x_0, r) \setminus \{x_0\}.$$

Then $L(c, P) \leq f(x_0) + g(x_0)$.

Proof. Relation to the paper's proof. The paper proves Theorem super-common by applying the Theorem on Sums (Lemma TOS, [CIL, Theorem 3.2]) through the argument of Proposition Lap-sup, producing jets $(\xi_\varepsilon + \nabla\varphi(x_\varepsilon), X_\varepsilon + \nabla^2\varphi(x_\varepsilon))$ of u and $(\eta_\varepsilon, Y_\varepsilon)$ of v with $\xi_\varepsilon = -\eta_\varepsilon$ and $X_\varepsilon + Y_\varepsilon \leq 0$, and then concluding by the chain (test-L). The sup-convolution/Jensen/Alexandrov steps below are exactly the standard proof of the Theorem on Sums (Crandall–Ishii–Lions, Theorem 3.2 and Appendix; the proof via sup-convolutions, Jensen's lemma and Alexandrov's theorem), specialized to the present situation of *common* variables and an operator independent of x : in this case the jets produced are genuine second-order superjets of the sup-convolutions at a single point y , which are transferred to u and v at nearby points $\hat{y}, \hat{y}'$ by `supConvolution_jet_transfer`; no closure of semijets is needed. The final step is the paper's chain (test-L) verbatim: ellipticity, then subadditivity, then the two subsolution inequalities, then continuity.

Notation. $K = \overline{B}(x_0, r)$ (compact, nonempty, $K \subseteq \Omega$); $w = u + v$; $m_0 = w(x_0) - q_0 = w(x_0) - Q(x_0)$. By clause (i) of `IsViscositySubsolution`, u, v are upper semicontinuous on Ω , hence on K . An upper semicontinuous function on a nonempty compact set is bounded above (it attains its maximum, by the argument in which a maximizing sequence has a convergent subsequence); fix $M_u \geq \sup_K u$, $M_v \geq \sup_K v$, and $M_Q \geq \sup_K |Q|$ (Q is continuous). Let $\|P\|$ be the operator norm, so $\langle Ph, h \rangle \leq \|P\| |h|^2$. Since P is symmetric, expanding gives for all y, h

$$(1) \quad Q(y + h) = Q(y) + \langle c + P(y - x_0), h \rangle + \frac{1}{2} \langle Ph, h \rangle.$$

For $\varepsilon > 0$ let $u^\varepsilon = u_K^\varepsilon, v^\varepsilon = v_K^\varepsilon$ (`SupConvolution`) and $W_\varepsilon = u^\varepsilon + v^\varepsilon - Q$. By `supConvolution_properties` (applied to u and to v on K): (i) $u(y) - \frac{|x-y|^2}{2\varepsilon} \leq u^\varepsilon(x)$ for $y \in K$; (ii) for each x there is $\hat{x} \in K$ with $u^\varepsilon(x) = u(\hat{x}) - \frac{|x-\hat{x}|^2}{2\varepsilon}$; (iii) $F_u^\varepsilon(x) = u^\varepsilon(x) + \frac{|x|^2}{2\varepsilon}$ is convex; and likewise for v . By (i) with $x = y = x_0$, $u^\varepsilon(x_0) \geq u(x_0)$ and $v^\varepsilon(x_0) \geq v(x_0)$, so

$$(2) \quad W_\varepsilon(x_0) \geq m_0.$$

Claim A (concentration). For every $\rho \in (0, r]$ there are $\beta_\rho > 0$ and $\varepsilon_\rho > 0$ such that $W_\varepsilon(y) \leq m_0 - \beta_\rho$ for all $\varepsilon \in (0, \varepsilon_\rho)$ and all $y \in A_\rho = \{y : \rho \leq |y - x_0| \leq r\}$.

Proof of Claim A. If $A_\rho = \emptyset$ any $\beta_\rho, \varepsilon_\rho > 0$ work. Otherwise $w - Q$ is upper semicontinuous on the compact nonempty set $A_\rho \subseteq K$ and attains its maximum m_ρ there at some $y_\rho \neq x_0$, so $m_\rho < m_0$ by the strict maximum hypothesis; let $\beta_\rho = \frac{1}{2}(m_0 - m_\rho) > 0$. Suppose no ε_ρ works. Then there are $\varepsilon_k \in (0, 1/k)$ and $y_k \in A_\rho$ with $W_{\varepsilon_k}(y_k) > m_0 - \beta_\rho$. Take $\hat{y}_k, \hat{y}'_k \in K$ as in (ii) for $u^{\varepsilon_k}(y_k)$ and $v^{\varepsilon_k}(y_k)$. Then

$$u(\hat{y}_k) + v(\hat{y}'_k) - Q(y_k) - \frac{|y_k - \hat{y}_k|^2 + |y_k - \hat{y}'_k|^2}{2\varepsilon_k} = W_{\varepsilon_k}(y_k) > m_0 - \beta_\rho.$$

Hence $u(\hat{y}_k) + v(\hat{y}'_k) - Q(y_k) > m_0 - \beta_\rho$ and $|y_k - \hat{y}_k|^2 + |y_k - \hat{y}'_k|^2 < 2\varepsilon_k C_0$ with $C_0 = M_u + M_v + M_Q - m_0 + \beta_\rho$. By compactness of A_ρ , along a subsequence $y_k \rightarrow y^* \in A_\rho$, and then $\hat{y}_k \rightarrow y^*, \hat{y}'_k \rightarrow y^*$ (as $\varepsilon_k \rightarrow 0$). Let $t > 0$. Upper semicontinuity of u and v on K at $y^* \in K$ (with $\hat{y}_k, \hat{y}'_k \in K$) and continuity of Q give, for large k , $u(\hat{y}_k) < u(y^*) + t$, $v(\hat{y}'_k) < v(y^*) + t$, $Q(y_k) > Q(y^*) - t$, so $m_0 - \beta_\rho < w(y^*) - Q(y^*) + 3t$. Letting $t \rightarrow 0$, $w(y^*) - Q(y^*) \geq m_0 - \beta_\rho = \frac{1}{2}(m_0 + m_\rho) > m_\rho$, contradicting the definition of m_ρ since $y^* \in A_\rho$. This proves Claim A.

Choice of parameters. Fix $\sigma > 0$. Since L is continuous on $\mathbb{R}^n \times \mathbb{S}^n$ at (c, P) , and f, g are continuous on Ω at $x_0 \in K \subseteq \Omega$, choose $\rho \in (0, r/2]$ such that

$$(3) \quad L(\xi, P) \geq L(c, P) - \sigma \text{ if } |\xi - c| \leq (\|P\| + 1)\rho, \quad f(y') \leq f(x_0) + \sigma, \quad g(y') \leq g(x_0) + \sigma \text{ if } y' \in \Omega, |y' - x_0| < 2\rho.$$

Let $\beta = \beta_\rho, \varepsilon_\rho$ be as in Claim A and $C_0 = M_u + M_v + M_Q - m_0 + \beta$. Choose $\varepsilon \in (0, \varepsilon_\rho)$ with $2\varepsilon \max(C_0, 0) < \rho^2$, and δ with $0 < \delta < \min(\rho, \frac{\beta}{2r})$.

Semiconvexity of W_ε . Let $\lambda = \|P\|$ and $C_W = \frac{2}{\varepsilon} + \lambda \geq 0$. Then

$$W_\varepsilon(x) + \frac{C_W}{2}|x|^2 = F_u^\varepsilon(x) + F_v^\varepsilon(x) + \left(\frac{\lambda}{2}|x|^2 - Q(x)\right).$$

The first two terms are convex by (iii). For the third, $\frac{\lambda}{2}|x|^2 = \frac{\lambda}{2}|x - x_0|^2 + \lambda\langle x_0, x - x_0 \rangle + \frac{\lambda}{2}|x_0|^2$, so $\frac{\lambda}{2}|x|^2 - Q(x) = \frac{1}{2}\langle (\lambda I - P)(x - x_0), x - x_0 \rangle + (\text{affine function of } x)$; the matrix $\lambda I - P$ is symmetric with $\langle (\lambda I - P)h, h \rangle \geq 0$, and a quadratic form $\kappa(h) = \frac{1}{2}\langle Gh, h \rangle$ with such G is convex

because $t\kappa(h_1) + (1-t)\kappa(h_2) - \kappa(th_1 + (1-t)h_2) = \frac{1}{2}t(1-t)\langle G(h_1 - h_2), h_1 - h_2 \rangle \geq 0$. Hence $W_\varepsilon + \frac{C_W}{2}|\cdot|^2$ is convex on $\mathbb{R}^n$ (a sum of convex functions).

Jensen. For $|y - x_0| = r$ we have $y \in A_\rho$ (as $\rho \leq r$), so by Claim A and (2), $W_\varepsilon(y) + 2\delta r \leq m_0 - \beta + 2\delta r < m_0 \leq W_\varepsilon(x_0)$. By `jensen_positive_measure` (with $W = W_\varepsilon$, $C = C_W$, $x_* = x_0$), the set K_δ of $y \in B(x_0, r)$ for which some p with $|p| \leq \delta$ satisfies $W_\varepsilon(y') + \langle p, y' \rangle \leq W_\varepsilon(y) + \langle p, y \rangle$ for all $y' \in K$ has $\text{vol}(K_\delta) > 0$.

Alexandrov. By `alexandrov_convex` applied to the convex functions F_u^ε and F_v^ε , there are null sets N_u, N_v outside of which these functions admit second-order expansions. Since $\text{vol}(K_\delta) \leq \text{vol}(K_\delta \setminus (N_u \cup N_v)) + \text{vol}(N_u) + \text{vol}(N_v)$, the set $K_\delta \setminus (N_u \cup N_v)$ has positive measure, so we can pick y in it, with a witness p , $|p| \leq \delta$. There are $a_u, b_v \in \mathbb{R}^n$ and symmetric A_u, B_v with $F_u^\varepsilon(y+h) - F_u^\varepsilon(y) - \langle a_u, h \rangle - \frac{1}{2}\langle A_u h, h \rangle = o(|h|^2)$ and similarly for F_v^ε with (b_v, B_v) . Since exactly $\frac{|y+h|^2}{2\varepsilon} = \frac{|y|^2}{2\varepsilon} + \langle \frac{y}{\varepsilon}, h \rangle + \frac{1}{2}\langle \frac{1}{\varepsilon}Ih, h \rangle$, putting $a = a_u - \frac{y}{\varepsilon}$, $A = A_u - \frac{1}{\varepsilon}I$, $b = b_v - \frac{y}{\varepsilon}$, $B = B_v - \frac{1}{\varepsilon}I$ (all A, B symmetric) we get, for every $\kappa > 0$ and all small $|h|$,

$$(4) \quad |u^\varepsilon(y+h) - u^\varepsilon(y) - \langle a, h \rangle - \frac{1}{2}\langle Ah, h \rangle| \leq \kappa|h|^2, \quad |v^\varepsilon(y+h) - v^\varepsilon(y) - \langle b, h \rangle - \frac{1}{2}\langle Bh, h \rangle| \leq \kappa|h|^2.$$

Location of $y, \hat{y}, \hat{y}'$. By the maximality defining K_δ (with $y' = x_0$) and (2), $W_\varepsilon(y) \geq W_\varepsilon(x_0) - \langle p, y - x_0 \rangle \geq m_0 - \delta r > m_0 - \beta$ (as $2\delta r < \beta$). By Claim A, $y \notin A_\rho$; since $y \in K$, $|y - x_0| < \rho$. Take $\hat{y}, \hat{y}' \in K$ as in (ii) for $u^\varepsilon(y), v^\varepsilon(y)$. Then

$$\frac{|y - \hat{y}|^2 + |y - \hat{y}'|^2}{2\varepsilon} = u(\hat{y}) + v(\hat{y}') - Q(y) - W_\varepsilon(y) \leq M_u + M_v + M_Q - m_0 + \beta = C_0,$$

so $|y - \hat{y}|, |y - \hat{y}'| \leq \sqrt{2\varepsilon \max(C_0, 0)} < \rho$, hence $|\hat{y} - x_0|, |\hat{y}' - x_0| < 2\rho \leq r$. Thus $\hat{y}, \hat{y}' \in B(x_0, r)$, which consists of interior points of K .

First- and second-order conditions at y . Since $y \in B(x_0, r)$, for all sufficiently small $|h|$ we have $y + h \in K$ and so $W_\varepsilon(y+h) + \langle p, h \rangle \leq W_\varepsilon(y)$. Put $\gamma = a + b - c - P(y - x_0) + p$ and $G = A + B - P$ (symmetric). By (1) and (4) with a fixed $\kappa > 0$, for all small $|h|$,

$$0 \geq W_\varepsilon(y+h) - W_\varepsilon(y) + \langle p, h \rangle \geq \langle \gamma, h \rangle + \frac{1}{2}\langle Gh, h \rangle - 2\kappa|h|^2.$$

For a unit vector e and small $t > 0$, taking $h = \pm te$ gives $\pm t\langle \gamma, e \rangle \leq (2\kappa + \frac{1}{2}\|G\|)t^2$; dividing by t and letting $t \rightarrow 0^+$ yields $\langle \gamma, e \rangle = 0$ for all e , so $\gamma = 0$, i.e.

$$a + b = c + P(y - x_0) - p.$$

Then with $h = te$: $\frac{1}{2}t^2\langle Ge, e \rangle \leq 2\kappa t^2$, so $\langle Ge, e \rangle \leq 4\kappa$; as $\kappa > 0$ is arbitrary, $\langle Ge, e \rangle \leq 0$ for all e . Hence $P - (A + B)$ is symmetric and positive semidefinite, i.e. $P \geq A + B$.

Subsolution inequalities via jet transfer. By (4), for every $\kappa > 0$, $u^\varepsilon(z) \leq u^\varepsilon(y) + \langle a, z - y \rangle + \frac{1}{2}\langle A(z - y), z - y \rangle + \kappa|z - y|^2$ for z near y ; moreover $u^\varepsilon(y) = u(\hat{y}) - \frac{|y - \hat{y}|^2}{2\varepsilon}$ with $\hat{y}$ interior to K , $K \subseteq \Omega$ compact, A symmetric. By `supConvolution_jet_transfer` (with $x = y$, $\hat{x} = \hat{y}$), $L(a, A) \leq f(\hat{y})$. In the same way $L(b, B) \leq g(\hat{y}')$.

The chain (test- L) and the limit. Let $\xi = c + P(y - x_0) - p = a + b$. Ellipticity (`IsDegenerateElliptic`, with $X = P \geq Y = A + B$, both symmetric), then subadditivity (`IsSubadditiveOperator`, A, B symmetric), then the two subsolution inequalities give

$$L(\xi, P) \leq L(\xi, A + B) = L(a + b, A + B) \leq L(a, A) + L(b, B) \leq f(\hat{y}) + g(\hat{y}').$$

Finally, $|\xi - c| \leq \|P\||y - x_0| + |p| \leq \|P\|\rho + \delta \leq (\|P\| + 1)\rho$ and $\hat{y}, \hat{y}' \in K \subseteq \Omega$ with $|\hat{y} - x_0|, |\hat{y}' - x_0| < 2\rho$, so by the continuity choices (3),

$$L(c, P) - \sigma \leq L(\xi, P) \leq f(\hat{y}) + g(\hat{y}') \leq f(x_0) + g(x_0) + 2\sigma.$$

Since $\sigma > 0$ was arbitrary, $L(c, P) \leq f(x_0) + g(x_0)$. □

3. MAIN THEOREMS OF THE PAPER

3.1. `infsuper_subsolution`.

Theorem 3.1 (Superposition in disjoint variables for the infinity Laplacian: subsolutions; paper Theorem infsuper). *Let $\mathcal{U} \subseteq \mathbb{R}^n$ and $\mathcal{V} \subseteq \mathbb{R}^m$ be domains (**IsOpenDomain**: open, nonempty, connected). Suppose $f \in C(\mathcal{U})$ and $g \in C(\mathcal{V})$ (i.e. $f : \mathbb{R}^n \rightarrow \mathbb{R}$ continuous on $\mathcal{U}$, $g : \mathbb{R}^m \rightarrow \mathbb{R}$ continuous on $\mathcal{V}$). Assume that $u : \mathbb{R}^n \rightarrow \mathbb{R}$ is a viscosity subsolution to*

$$-\Delta_\infty u(x) = f(x) \quad \text{in } \mathcal{U}$$

and $v : \mathbb{R}^m \rightarrow \mathbb{R}$ is a viscosity subsolution to

$$-\Delta_\infty v(y) = g(y) \quad \text{in } \mathcal{V},$$

*in the sense of **IsViscositySubsolution** with the operator $L_\infty(\xi, X) = -\langle X\xi, \xi \rangle$ of **InfLaplaceOperator** (so $L_\infty(\nabla\psi, \nabla^2\psi) = -\Delta_\infty\psi$). Then $w(x, y) = u(x) + v(y)$ is a viscosity subsolution to*

$$-\Delta_\infty w(x, y) = f(x) + g(y) \quad \text{in } \mathcal{U} \times \mathcal{V} \subseteq \mathbb{R}^n \times \mathbb{R}^m.$$

*Here $\mathbb{R}^n \times \mathbb{R}^m$ is identified with $\mathbb{R}^{n+m}$ by concatenating coordinates, $(x, y) \leftrightarrow z$ with $x = \text{fst } z$ (**EuclidFst**) and $y = \text{snd } z$ (**EuclidSnd**); thus $\mathcal{U} \times \mathcal{V}$ is the set $\{z \in \mathbb{R}^{n+m} : \text{fst } z \in \mathcal{U}, \text{snd } z \in \mathcal{V}\}$, $w(z) = u(\text{fst } z) + v(\text{snd } z)$, and the right-hand side is $z \mapsto f(\text{fst } z) + g(\text{snd } z)$.*

Proof. Write $\Omega = \{z : \text{fst } z \in \mathcal{U}, \text{snd } z \in \mathcal{V}\}$, $w(z) = u(\text{fst } z) + v(\text{snd } z)$, $F(z) = f(\text{fst } z) + g(\text{snd } z)$. The block maps fst, snd are linear (coordinate selections), hence continuous. $\mathcal{U}, \mathcal{V}$ are open (domains), so $\Omega = \text{fst}^{-1}(\mathcal{U}) \cap \text{snd}^{-1}(\mathcal{V})$ is open. We verify the two clauses of **IsViscositySubsolution** for w on Ω with right-hand side F .

(i) *Upper semicontinuity.* Let $z \in \Omega$ and $t > w(z)$, $s = \frac{1}{2}(t - w(z)) > 0$. By clause (i) for u , there is a neighbourhood N_1 of $\text{fst } z$ with $u < u(\text{fst } z) + s$ on $N_1 \cap \mathcal{U}$; by clause (i) for v , a neighbourhood N_2 of $\text{snd } z$ with $v < v(\text{snd } z) + s$ on $N_2 \cap \mathcal{V}$. Then $N = \text{fst}^{-1}(N_1) \cap \text{snd}^{-1}(N_2)$ is a neighbourhood of z and for $z' \in N \cap \Omega$ we have $\text{fst } z' \in N_1 \cap \mathcal{U}$, $\text{snd } z' \in N_2 \cap \mathcal{V}$, so $w(z') < w(z) + 2s = t$.

(ii) *Test functions.* Let $z_0 \in \Omega$ and ψ be C^2 on Ω with $w - \psi$ having a local maximum at z_0 . Write $c = \nabla\psi(z_0)$ and $H = \nabla^2\psi(z_0)$ (**EuclidHessian**). Fix $\eta > 0$. By **c2_local_max_to_strict_quadratic** (in dimension $n + m$, with the open set Ω , ψ , w , z_0 , η), H is symmetric and there is $r > 0$ with $\overline{B}(z_0, r) \subseteq \Omega$ and, with $Q_\eta(z) = \psi(z_0) + \langle c, z - z_0 \rangle + \frac{1}{2}\langle (H + 2\eta I)(z - z_0), z - z_0 \rangle$,

$$w(z) - Q_\eta(z) \leq w(z_0) - \psi(z_0) - \frac{\eta}{2}|z - z_0|^2 < w(z_0) - \psi(z_0) \quad (z \in \overline{B}(z_0, r) \setminus \{z_0\}).$$

So the hypotheses of **disjoint_sum_quadratic_strict** hold with $q_0 = \psi(z_0)$, c , and the symmetric matrix $P = H + 2\eta I$ (together with continuity of f on $\mathcal{U}$, of g on $\mathcal{V}$, and the subsolution properties of u, v). It gives

$$L_\infty(c, H + 2\eta I) \leq f(\text{fst } z_0) + g(\text{snd } z_0) = F(z_0).$$

Since $L_\infty(c, H + 2\eta I) = -\langle Hc, c \rangle - 2\eta|c|^2 = L_\infty(c, H) - 2\eta|c|^2$, we get $L_\infty(c, H) \leq F(z_0) + 2\eta|c|^2$ for every $\eta > 0$. Letting $\eta \rightarrow 0^+$, $L_\infty(\nabla\psi(z_0), \nabla^2\psi(z_0)) \leq F(z_0)$, which is clause (ii). Hence w is a viscosity subsolution of $-\Delta_\infty w = F$ in Ω . $\square$

3.2. infsuper_supersolution.

Theorem 3.2 (Superposition in disjoint variables for the infinity Laplacian: supersolutions; paper Theorem infsuper). *Let $\mathcal{U} \subseteq \mathbb{R}^n$ and $\mathcal{V} \subseteq \mathbb{R}^m$ be domains (**IsOpenDomain**). Suppose $f \in C(\mathcal{U})$ and $g \in C(\mathcal{V})$. Assume that $u : \mathbb{R}^n \rightarrow \mathbb{R}$ is a viscosity supersolution to $-\Delta_\infty u(x) = f(x)$ in $\mathcal{U}$ and $v : \mathbb{R}^m \rightarrow \mathbb{R}$ is a viscosity supersolution to $-\Delta_\infty v(y) = g(y)$ in $\mathcal{V}$, in the sense of **IsViscositySupersolution** with the operator $L_\infty(\xi, X) = -\langle X\xi, \xi \rangle$ of **InfLaplaceOperator**. Then $w(x, y) = u(x) + v(y)$ is a viscosity supersolution to*

$$-\Delta_\infty w(x, y) = f(x) + g(y) \quad \text{in } \mathcal{U} \times \mathcal{V} \subseteq \mathbb{R}^n \times \mathbb{R}^m,$$

*where $\mathbb{R}^n \times \mathbb{R}^m$ is identified with $\mathbb{R}^{n+m}$ by concatenating coordinates ($x = \text{fst } z$ from **EuclidFst**, $y = \text{snd } z$ from **EuclidSnd**), $\mathcal{U} \times \mathcal{V} = \{z : \text{fst } z \in \mathcal{U}, \text{snd } z \in \mathcal{V}\}$, $w(z) = u(\text{fst } z) + v(\text{snd } z)$, and the right-hand side is $f(\text{fst } z) + g(\text{snd } z)$.*

Proof. This is the paper's remark that the argument also applies to supersolutions, realized by negation. Let $\Omega = \{z : \text{fst } z \in \mathcal{U}, \text{snd } z \in \mathcal{V}\}$.

By `infLaplace_supersolution_iff_neg` ($\Rightarrow$, applied in $\mathbb{R}^n$ to $u, f, \mathcal{U}$ and in $\mathbb{R}^m$ to $v, g, \mathcal{V}$), $-u$ is a viscosity subsolution of $L_\infty = -f$ in $\mathcal{U}$ and $-v$ is one of $L_\infty = -g$ in $\mathcal{V}$. The functions $-f, -g$ are continuous on $\mathcal{U}, \mathcal{V}$. By `infsuper_subsolution` (applied to $-f, -g, -u, -v$), the function $z \mapsto (-u)(\text{fst } z) + (-v)(\text{snd } z) = -w(z)$ is a viscosity subsolution in Ω of $L_\infty = z \mapsto (-f)(\text{fst } z) + (-g)(\text{snd } z) = -F(z)$, where $F(z) = f(\text{fst } z) + g(\text{snd } z)$ (pointwise identities of functions). By `infLaplace_supersolution_iff_neg` ($\Leftarrow$, in $\mathbb{R}^{n+m}$ with Ω, F and w), w is a viscosity supersolution of $L_\infty(\nabla w, \nabla^2 w) = F$, i.e. of $-\Delta_\infty w = f(x) + g(y)$, in Ω . $\square$

3.3. `infsuper_solution`.

Theorem 3.3 (Superposition in disjoint variables for the infinity Laplacian: solutions; paper Theorem `infsuper`). *Let $\mathcal{U} \subseteq \mathbb{R}^n$ and $\mathcal{V} \subseteq \mathbb{R}^m$ be domains (`IsOpenDomain`). Suppose $f \in C(\mathcal{U})$ and $g \in C(\mathcal{V})$. Assume that $u : \mathbb{R}^n \rightarrow \mathbb{R}$ is a viscosity solution to $-\Delta_\infty u(x) = f(x)$ in $\mathcal{U}$ and $v : \mathbb{R}^m \rightarrow \mathbb{R}$ is a viscosity solution to $-\Delta_\infty v(y) = g(y)$ in $\mathcal{V}$, in the sense of `IsViscositySolution` with the operator $L_\infty(\xi, X) = -\langle X\xi, \xi \rangle$ of `InfLaplaceOperator`. Then $w(x, y) = u(x) + v(y)$ is a viscosity solution to*

$$-\Delta_\infty w(x, y) = f(x) + g(y) \quad \text{in } \mathcal{U} \times \mathcal{V} \subseteq \mathbb{R}^n \times \mathbb{R}^m,$$

with $\mathcal{U} \times \mathcal{V} = \{z \in \mathbb{R}^{n+m} : \text{fst } z \in \mathcal{U}, \text{snd } z \in \mathcal{V}\}$, $w(z) = u(\text{fst } z) + v(\text{snd } z)$ and right-hand side $f(\text{fst } z) + g(\text{snd } z)$, where $\mathbb{R}^n \times \mathbb{R}^m$ is identified with $\mathbb{R}^{n+m}$ by concatenating coordinates ($x = \text{fst } z$ from `EuclidFst`, $y = \text{snd } z$ from `EuclidSnd`).

Proof. By `IsViscositySolution`, u is both a viscosity subsolution and a viscosity supersolution of $-\Delta_\infty u = f$ in $\mathcal{U}$, and likewise v for g in $\mathcal{V}$. By `infsuper_subsolution` (applied to the subsolution parts), w is a viscosity subsolution of $-\Delta_\infty w = f(x) + g(y)$ in $\mathcal{U} \times \mathcal{V}$; by `infsuper_supersolution` (applied to the supersolution parts), w is a viscosity supersolution of the same equation there. By `IsViscositySolution`, w is a viscosity solution. $\square$

3.4. `superposition_common_subsolution`.

Theorem 3.4 (Subsolutions of sub-additive equations; paper Theorem `super-common`). *Let $\Omega \subseteq \mathbb{R}^n$ be a domain (`IsOpenDomain`: open, nonempty and connected), and let $f, g : \mathbb{R}^n \rightarrow \mathbb{R}$ be continuous on Ω , i.e. $f, g \in C(\Omega)$. Let $L : \mathbb{R}^n \times \mathbb{R}^{n \times n} \rightarrow \mathbb{R}$ be continuous on $\mathbb{R}^n \times \mathbb{S}^n$ ($\mathbb{S}^n$ the symmetric real $n \times n$ matrices), elliptic in the sense of (ellip) (`IsDegenerateElliptic`: $L(\xi, X) \leq L(\xi, Y)$ whenever $X, Y \in \mathbb{S}^n$, $X \geq Y$), and subadditive in the sense of (sub-add) (`IsSubadditiveOperator`: $L(\xi + \eta, X + Y) \leq L(\xi, X) + L(\eta, Y)$ for $\xi, \eta \in \mathbb{R}^n$, $X, Y \in \mathbb{S}^n$). Assume that $u : \mathbb{R}^n \rightarrow \mathbb{R}$, upper semicontinuous on Ω , is a viscosity subsolution of*

$$L(\nabla u, \nabla^2 u) = f \quad \text{in } \Omega,$$

and $v : \mathbb{R}^n \rightarrow \mathbb{R}$, upper semicontinuous on Ω , is a viscosity subsolution of

$$L(\nabla v, \nabla^2 v) = g \quad \text{in } \Omega$$

(both in the sense of `IsViscositySubsolution`, which includes upper semicontinuity on Ω). Then $w = u + v$ is a viscosity subsolution of

$$L(\nabla w, \nabla^2 w) = f + g \quad \text{in } \Omega.$$

Proof. Write $w = u + v$. We verify the two clauses of `IsViscositySubsolution` for w with right-hand side $f + g$.

(i) *Upper semicontinuity.* Let $x \in \Omega$ and $t > w(x)$; put $s = \frac{1}{2}(t - w(x)) > 0$. By upper semicontinuity of u and v on Ω (clause (i) for u and v) there is a neighbourhood U of x with $u < u(x) + s$ and $v < v(x) + s$ on $U \cap \Omega$; then $w < w(x) + 2s = t$ on $U \cap \Omega$. So w is upper semicontinuous on Ω .

(ii) *Test functions.* Let $x_0 \in \Omega$ and let $\psi : \mathbb{R}^n \rightarrow \mathbb{R}$ be C^2 on Ω with $w - \psi$ attaining a local maximum at x_0 . Write $H = \nabla^2 \psi(x_0)$ (`EuclidHessian`). Ω is open since it is a domain. Fix

$\eta > 0$. By `c2_local_max_to_strict_quadratic` (with this Ω , ψ , w , x_0 , η), H is symmetric and there is $r > 0$ with $\overline{B}(x_0, r) \subseteq \Omega$ and

$$w(y) - Q_\eta(y) \leq w(x_0) - \psi(x_0) - \frac{\eta}{2}|y - x_0|^2 \quad (y \in \overline{B}(x_0, r)), \quad Q_\eta(y) = \psi(x_0) + \langle \nabla \psi(x_0), y - x_0 \rangle + \frac{1}{2} \langle (H + 2\eta I)(y - x_0), y - x_0 \rangle$$

For $y \in \overline{B}(x_0, r) \setminus \{x_0\}$ the right-hand side is $< w(x_0) - \psi(x_0)$. Hence the hypotheses of `sum_subsolution_quadratic_strict` hold with $q_0 = \psi(x_0)$, $c = \nabla \psi(x_0)$ and the symmetric matrix $P = H + 2\eta I$ (the remaining hypotheses — continuity of f, g on Ω , continuity of L on $\mathbb{R}^n \times \mathbb{S}^n$, ellipticity, subadditivity, and the subsolution properties of u and v — are hypotheses of the theorem). It gives

$$L(\nabla \psi(x_0), H + 2\eta I) \leq f(x_0) + g(x_0) \quad \text{for every } \eta > 0.$$

As $\eta \rightarrow 0^+$, $(\nabla \psi(x_0), H + 2\eta I) \rightarrow (\nabla \psi(x_0), H)$ within $\mathbb{R}^n \times \mathbb{S}^n$ (H and $H + 2\eta I$ are symmetric), so by continuity of L on $\mathbb{R}^n \times \mathbb{S}^n$,

$$L(\nabla \psi(x_0), \nabla^2 \psi(x_0)) \leq f(x_0) + g(x_0) = (f + g)(x_0).$$

This is clause (ii) for w , so $w = u + v$ is a viscosity subsolution of $L(\nabla w, \nabla^2 w) = f + g$ in Ω . $\square$

3.5. superposition_cone_above.

Theorem 3.5 (Superposition for comparison with cones from above). *Let $(\mathbf{X}, d_{\mathbf{X}})$ and $(\mathbf{Y}, d_{\mathbf{Y}})$ be proper geodesic spaces (`IsGeodesicSpace`), and let $\mathcal{U} \subsetneq \mathbf{X}$ and $\mathcal{V} \subsetneq \mathbf{Y}$ be domains (`IsOpenDomain`). Equip $\mathbf{X} \times \mathbf{Y}$ with the ℓ^1 product metric*

$$d_1((x_1, y_1), (x_2, y_2)) = d_{\mathbf{X}}(x_1, x_2) + d_{\mathbf{Y}}(y_1, y_2),$$

and let $\mathcal{W} = \mathcal{U} \times \mathcal{V}$. Let $u : \mathbf{X} \rightarrow \mathbb{R}$ and $v : \mathbf{Y} \rightarrow \mathbb{R}$ be such that u is continuous on $\mathcal{U}$, v is continuous on $\mathcal{V}$, u satisfies comparison with cones from above in $\mathcal{U}$, and v satisfies comparison with cones from above in $\mathcal{V}$ (`ConeComparisonAbove`). Then the disjoint sum

$$w(x, y) = u(x) + v(y)$$

is continuous on $\mathcal{W}$ and satisfies comparison with cones from above in $\mathcal{W}$ with respect to the metric d_1 .

Proof. Write $\pi_1(x, y) = x$, $\pi_2(x, y) = y$, and note $d_1(z, z') = d_{\mathbf{X}}(\pi_1 z, \pi_1 z') + d_{\mathbf{Y}}(\pi_2 z, \pi_2 z')$.

Step 0: continuity and local upper bound. By `ConeComparisonAbove` (clause 1) applied to u in $\mathcal{U}$ and to v in $\mathcal{V}$, u is locally bounded from above in $\mathcal{U}$ and v is locally bounded from above in $\mathcal{V}$. Together with continuity of u on $\mathcal{U}$ and of v on $\mathcal{V}$, `disjointSum_continuousOn_locallyBddAbove` gives that w is continuous on $\mathcal{W}$ and locally bounded from above in $\mathcal{W}$. This proves the first conclusion and clause 1 of `ConeComparisonAbove` for w in $\mathcal{W}$.

Step 1: setting up clause 2. Let $\hat{z} = (\hat{x}, \hat{y}) \in \mathcal{W}$, $c \in \mathbb{R}$, $\kappa \geq 0$, and let $\mathcal{O} \subseteq \mathbf{X} \times \mathbf{Y}$ be open and bounded with $\overline{\mathcal{O}}$ compact, $\overline{\mathcal{O}} \subseteq \mathcal{W}$, and $\hat{z} \notin \mathcal{O}$. Let $\phi(z) = c + \kappa d_1(\hat{z}, z) = c + \kappa d_{\mathbf{X}}(\hat{x}, \pi_1 z) + \kappa d_{\mathbf{Y}}(\hat{y}, \pi_2 z)$ and assume

$$(1) \quad w \leq \phi \quad \text{on } \partial \mathcal{O}.$$

Let $z_0 = (x_0, y_0) \in \overline{\mathcal{O}}$; we must show $w(z_0) \leq \phi(z_0)$. Suppose, for contradiction, that

$$(2) \quad w(z_0) > \phi(z_0).$$

Then $z_0 \in \mathcal{O}$: otherwise $z_0 \in \overline{\mathcal{O}} \setminus \mathcal{O} = \overline{\mathcal{O}} \setminus \text{int } \mathcal{O} = \partial \mathcal{O}$ ($\mathcal{O}$ is open), and (1) contradicts (2). (Unlike the paper we do not need z_0 to be a maximizer of $w - \phi$; only (2) is used.)

Step 2: the $\mathbf{Y}$ -slice. Let $\mathcal{O}_{\mathbf{Y}} = \{y : (\hat{x}, y) \in \mathcal{O}\}$. By `slice_right_admissible` (with $\mathcal{U}, \mathcal{V}, \mathcal{O}, \hat{x}$), $\mathcal{O}_{\mathbf{Y}}$ is open and bounded, $\overline{\mathcal{O}_{\mathbf{Y}}}$ is compact, $\overline{\mathcal{O}_{\mathbf{Y}}} \subseteq \mathcal{V}$, and $\partial \mathcal{O}_{\mathbf{Y}} \subseteq \{y : (\hat{x}, y) \in \partial \mathcal{O}\}$. Moreover $\hat{y} \in \mathcal{V}$ and $\hat{y} \notin \mathcal{O}_{\mathbf{Y}}$, because $(\hat{x}, \hat{y}) = \hat{z} \notin \mathcal{O}$. Consider the cone $\phi_{\mathbf{Y}} = (c - u(\hat{x})) + \kappa d_{\mathbf{Y}}(\hat{y}, \cdot)$ with vertex $\hat{y}$ and slope $\kappa \geq 0$. For $y \in \partial \mathcal{O}_{\mathbf{Y}}$ we have $(\hat{x}, y) \in \partial \mathcal{O}$, so by (1)

$$u(\hat{x}) + v(y) = w(\hat{x}, y) \leq c + \kappa d_{\mathbf{X}}(\hat{x}, \hat{x}) + \kappa d_{\mathbf{Y}}(\hat{y}, y) = c + \kappa d_{\mathbf{Y}}(\hat{y}, y),$$

i.e. $v \leq \phi_{\mathbf{Y}}$ on $\partial \mathcal{O}_{\mathbf{Y}}$. Clause 2 of `ConeComparisonAbove` for v in $\mathcal{V}$ therefore yields

$$(3) \quad u(\hat{x}) + v(y) \leq c + \kappa d_{\mathbf{Y}}(\hat{y}, y) = \phi(\hat{x}, y) \quad \text{for all } y \in \overline{\mathcal{O}_{\mathbf{Y}}}.$$

Step 3: $x_0 \neq \hat{x}$. If $x_0 = \hat{x}$, then $(\hat{x}, y_0) = z_0 \in \mathcal{O}$, so $y_0 \in \mathcal{O}_{\mathbf{Y}} \subseteq \overline{\mathcal{O}_{\mathbf{Y}}}$ and (3) gives $w(z_0) = u(\hat{x}) + v(y_0) \leq c + \kappa d_{\mathbf{Y}}(\hat{y}, y_0) = c + \kappa d_{\mathbf{X}}(\hat{x}, x_0) + \kappa d_{\mathbf{Y}}(\hat{y}, y_0) = \phi(z_0)$, contradicting (2).

Step 4: the $\mathbf{X}$ -slice. Let $\mathcal{O}_{\mathbf{X}} = \{x : (x, y_0) \in \mathcal{O}\} \setminus \{\hat{x}\}$. By `slice_left_admissible` (with $\mathcal{U}, \mathcal{V}, \mathcal{O}, y_0, \hat{x}$), $\mathcal{O}_{\mathbf{X}}$ is open and bounded, $\overline{\mathcal{O}_{\mathbf{X}}}$ is compact, $\overline{\mathcal{O}_{\mathbf{X}}} \subseteq \mathcal{U}$, $\overline{\mathcal{O}_{\mathbf{X}}} \subseteq \{x : (x, y_0) \in \overline{\mathcal{O}}\}$, and $\partial \mathcal{O}_{\mathbf{X}} \subseteq \{x : (x, y_0) \in \partial \mathcal{O}\} \cup \{\hat{x}\}$. Clearly $\hat{x} \notin \mathcal{O}_{\mathbf{X}}$, and $\hat{x} \in \mathcal{U}$. Consider the cone

$$\phi_{\mathbf{X}}(x) = (c + \kappa d_{\mathbf{Y}}(\hat{y}, y_0) - v(y_0)) + \kappa d_{\mathbf{X}}(\hat{x}, x)$$

with vertex $\hat{x}$ and slope $\kappa \geq 0$; note $u(x) \leq \phi_{\mathbf{X}}(x)$ is equivalent to $w(x, y_0) \leq \phi(x, y_0)$. We check $u \leq \phi_{\mathbf{X}}$ on $\partial \mathcal{O}_{\mathbf{X}}$. Let $x \in \partial \mathcal{O}_{\mathbf{X}}$.

- If $(x, y_0) \in \partial \mathcal{O}$, then (1) gives $u(x) + v(y_0) \leq c + \kappa d_{\mathbf{X}}(\hat{x}, x) + \kappa d_{\mathbf{Y}}(\hat{y}, y_0)$, i.e. $u(x) \leq \phi_{\mathbf{X}}(x)$.
- Otherwise $x = \hat{x}$, and we must show $u(\hat{x}) \leq c + \kappa d_{\mathbf{Y}}(\hat{y}, y_0) - v(y_0)$ (as $d_{\mathbf{X}}(\hat{x}, \hat{x}) = 0$). If $(\hat{x}, y_0) \in \mathcal{O}$, then $y_0 \in \mathcal{O}_{\mathbf{Y}} \subseteq \overline{\mathcal{O}_{\mathbf{Y}}}$ and (3) with $y = y_0$ gives exactly this. If $(\hat{x}, y_0) \notin \mathcal{O}$, then, since $\hat{x} = x \in \partial \mathcal{O}_{\mathbf{X}} \subseteq \overline{\mathcal{O}_{\mathbf{X}}}$, we have $(\hat{x}, y_0) \in \overline{\mathcal{O}} \setminus \mathcal{O} = \partial \mathcal{O}$, and (1) gives $u(\hat{x}) + v(y_0) \leq c + 0 + \kappa d_{\mathbf{Y}}(\hat{y}, y_0)$, which is again the desired inequality.

Hence clause 2 of `ConeComparisonAbove` for u in $\mathcal{U}$ (with vertex $\hat{x} \in \mathcal{U} \setminus \mathcal{O}_{\mathbf{X}}$, constant $c + \kappa d_{\mathbf{Y}}(\hat{y}, y_0) - v(y_0)$, slope κ , set $\mathcal{O}_{\mathbf{X}}$) yields $u \leq \phi_{\mathbf{X}}$ on $\overline{\mathcal{O}_{\mathbf{X}}}$. Now $(x_0, y_0) = z_0 \in \mathcal{O}$ and $x_0 \neq \hat{x}$ by Step 3, so $x_0 \in \mathcal{O}_{\mathbf{X}} \subseteq \overline{\mathcal{O}_{\mathbf{X}}}$ and

$$w(z_0) = u(x_0) + v(y_0) \leq c + \kappa d_{\mathbf{X}}(\hat{x}, x_0) + \kappa d_{\mathbf{Y}}(\hat{y}, y_0) = \phi(z_0),$$

contradicting (2). Hence $w \leq \phi$ on $\overline{\mathcal{O}}$, which is clause 2 of `ConeComparisonAbove` for w in $\mathcal{W}$ with respect to d_1 .

The hypotheses that $\mathbf{X}, \mathbf{Y}$ are proper and geodesic and that $\mathcal{U}, \mathcal{V}$ are proper subdomains are the paper's standing setting for the definition of comparison with cones; the argument above does not otherwise use them. $\square$

3.6. `superposition_cone_below`.

Corollary 3.6 (Superposition for comparison with cones from below). *Let $(\mathbf{X}, d_{\mathbf{X}})$ and $(\mathbf{Y}, d_{\mathbf{Y}})$ be proper geodesic spaces (`IsGeodesicSpace`), and let $\mathcal{U} \subsetneq \mathbf{X}$ and $\mathcal{V} \subsetneq \mathbf{Y}$ be domains (`IsOpenDomain`). Equip $\mathbf{X} \times \mathbf{Y}$ with the ℓ^1 product metric $d_1((x_1, y_1), (x_2, y_2)) = d_{\mathbf{X}}(x_1, x_2) + d_{\mathbf{Y}}(y_1, y_2)$ and let $\mathcal{W} = \mathcal{U} \times \mathcal{V}$. Let $u : \mathbf{X} \rightarrow \mathbb{R}$ and $v : \mathbf{Y} \rightarrow \mathbb{R}$ be such that u is continuous on $\mathcal{U}$, v is continuous on $\mathcal{V}$, u satisfies comparison with cones from below in $\mathcal{U}$, and v satisfies comparison with cones from below in $\mathcal{V}$ (`ConeComparisonBelow`). Then $w(x, y) = u(x) + v(y)$ is continuous on $\mathcal{W}$ and satisfies comparison with cones from below in $\mathcal{W}$ with respect to d_1 .*

Proof. By `ConeComparisonBelow`, the hypotheses say that $-u$ satisfies comparison with cones from above in $\mathcal{U}$ and $-v$ satisfies comparison with cones from above in $\mathcal{V}$ (`ConeComparisonAbove`). Also $-u$ is continuous on $\mathcal{U}$ and $-v$ is continuous on $\mathcal{V}$. All remaining hypotheses (proper geodesic spaces, proper subdomains) are unchanged, so `superposition_cone_above` applied to $-u, -v$ shows that $\tilde{w}(x, y) = (-u(x)) + (-v(y))$ is continuous on $\mathcal{W}$ and satisfies comparison with cones from above in $\mathcal{W}$ with respect to d_1 . Since $\tilde{w} = -w$ pointwise, $w = -\tilde{w}$ is continuous on $\mathcal{W}$, and $-w = \tilde{w}$ satisfies comparison with cones from above in $\mathcal{W}$, which by `ConeComparisonBelow` means that w satisfies comparison with cones from below in $\mathcal{W}$. $\square$

3.7. `superposition_cone_both`.

Corollary 3.7 (Superposition for comparison with cones from both sides). *Let $(\mathbf{X}, d_{\mathbf{X}})$ and $(\mathbf{Y}, d_{\mathbf{Y}})$ be proper geodesic spaces (`IsGeodesicSpace`), and let $\mathcal{U} \subsetneq \mathbf{X}$ and $\mathcal{V} \subsetneq \mathbf{Y}$ be domains (`IsOpenDomain`). Equip $\mathbf{X} \times \mathbf{Y}$ with the ℓ^1 product metric $d_1((x_1, y_1), (x_2, y_2)) = d_{\mathbf{X}}(x_1, x_2) + d_{\mathbf{Y}}(y_1, y_2)$ and let $\mathcal{W} = \mathcal{U} \times \mathcal{V}$. Let $u : \mathbf{X} \rightarrow \mathbb{R}$ and $v : \mathbf{Y} \rightarrow \mathbb{R}$ be such that u is continuous on $\mathcal{U}$, v is continuous on $\mathcal{V}$, u satisfies comparison with cones (from both sides) in $\mathcal{U}$, and v satisfies comparison with cones (from both sides) in $\mathcal{V}$ (`ConeComparisonBoth`). Then $w(x, y) = u(x) + v(y)$ is continuous on $\mathcal{W}$ and satisfies comparison with cones (from both sides) in $\mathcal{W}$ with respect to d_1 .*

Proof. By `ConeComparisonBoth`, u satisfies comparison with cones from above and from below in $\mathcal{U}$, and likewise v in $\mathcal{V}$. Applying `superposition_cone_above` to u, v gives that w is continuous on $\mathcal{W}$ and satisfies comparison with cones from above in $\mathcal{W}$. Applying `superposition_cone_below` to u, v gives that w satisfies comparison with cones from below in $\mathcal{W}$. By `ConeComparisonBoth`, w satisfies comparison with cones from both sides in $\mathcal{W}$ with respect to d_1 . $\square$